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Review

Artificial neural networks applied to olive oil production and characterization: A systematic review

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ABSTRACT

In recent years, the olive oil sector has experienced growth due to the health benefits associated with olive oil and its increasing demand in international markets. Artificial Neural Networks (ANNs) have emerged as powerful tools in various scientific domains, enhancing both the efficiency and the accuracy of analyses in the olive oil sector. This paper aims to comprehensively review the adoption of ANNs in the assessment of olive oil across production and post-production stages. To achieve this goal, we followed the well-known guidelines of Kitchenham (2004) for performing *systematic reviews*. This up-to-date review examines literature from the last seven years, analyzing 628 publications and finally selecting 79 primary studies. Through a systematic and comprehensive analysis, this review seeks to provide insights into the current state of research, identify gaps in knowledge, and offer recommendations for future directions in harnessing ANNs to optimize the production and post-production analyses of olive oil.

1. Introduction

The olive oil industry plays a crucial role in the global economy, contributing significantly to the food and agricultural sectors. The popularity of olive oil stems not only from its diverse culinary applications but also from its associated health benefits, including anti-inflammatory and antioxidant properties. As such, quality control, authenticity verification, and accurate characterization of olive oil are of paramount importance, both from an economic and a health perspective.

Traditionally, the production, characterization, and authentication of olive oil have relied on physical and chemical analytical methods, coupled with sensory evaluation. These conventional methods, though reliable, are often resource-intensive, time-consuming, and require skilled technicians. In addition, they may not effectively capture the complexity and subtlety of variables involved in olive oil production and characterization.

In recent years, machine learning has experienced a significant surge in popularity. This increase is notable not only among the general public, thanks to advancements in large language models like ChatGPT, but also within the scientific community. Artificial Neural Networks (ANNs), inspired by biological neural networks, are computational models capable of learning from and interpreting complex patterns within data. ANNs' abilities in pattern recognition, classification, and prediction have proven to be remarkably beneficial across various industrial sectors (Meireles, Almeida, & Simoes, 2003). Despite ANNs being accessible for many years through toolboxes and packages in major statistical software suites, their adoption and exploration in research have intensified only recently.

In the olive oil industry, the application of ANNs holds considerable potential in enhancing production efficiency, improving oil characterization, and helping authentication processes. By learning from vast datasets on olive oil's physicochemical properties, ANNs can potentially discern intricate patterns, make accurate predictions, and detect anomalies, including adulteration or incorrect classification.

ANNs can significantly improve several pain points of the olive oil industry by enhancing decision-making, optimizing processes, and reducing costs. Among them, we could underline: (i) yield prediction in which predictive models combining temperature, humidity, and olive ripeness data could recommend the best harvest time to increase the olive yield and improve the oil quality; (ii) quality control of olive oil in which ANN models can be used to detect adulteration, perform cultivar discrimination or olive oil classification; (iii) olive oil quality prediction, where ANNs can be used to determine factors like fatty acid composition, oxidative stability levels, and polyphenol content. The goal of this paper is to analyze the literature focusing solely on aspects related to olive oil production and characterization.

As shown in Section 3, several literature reviews deal with the adoption of ANNs for the production and characterization of olive oil. Some

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of them consider the general problem of analytical evaluation of food (including olive oil) but are not focused on olive oil (Esteki et al., 2018; Sanaeifar, ZakiDizaji, Jafari, & de la Guardia, 2017), others address only a single goal of using ANNs, such as adulteration (Meenu, Cai, & Xu, 2019; Natarajan & Ponnusamy, 2022) or shelf life (Martín-Torres, Ruiz-Castro, Jiménez-Carvelo, & Cuadros-Rodríguez, 2022), others are mainly focused on the ANN input type such as spectroscopy methods (Ordoudi, Strani, & Cocchi, 2023) or e-nose (Sanaeifar et al., 2017). Other works informally review ANN or machine learning methods, e.g., Gonzalez-Fern et al. (2019), but do not examine the literature systematically.

The contribution of this paper is a systematic and up-to-date review of the application of ANNs in the olive oil sector over the past seven years. Following established systematic review guidelines (Kitchenham, 2004), we analyzed 570 publications from six digital libraries, enhanced with 58 additional studies identified through snowballing over Google Scholar. After a rigorous selection process, 79 primary studies were included in our review. We categorize and evaluate the objectives of ANN applications in olive oil processing, the types of ANNs used, their input variables, and the performance metrics adopted.

This study highlights the potential of ANNs as a valuable research tool in the olive oil sector. ANNs have been employed for various purposes, including adulteration detection, classification, geographical discrimination, attribute prediction, and optimization of production processes. Their ability to process diverse data types enables them to outperform traditional chemometric methods in handling complex datasets. According to the results of our study, Multi-Layer Perceptron (MLP) is the most frequently ANN architecture used for evaluating oxidative stability, while both MLPs and Convolutional Neural Networks (CNNs) are widely applied in predicting olive oil characteristics. Spectroscopy has emerged as a key data source, particularly for adulteration detection. Furthermore, this review highlights common pitfalls of the application of ANN to the olive oil sector, such as overfitting and low availability of large datasets. Finally, we outline future research opportunities to further integrate ANNs with emerging technologies to improve the quality, authenticity, and efficiency of the olive oil industry.

The remainder of the article is structured as follows: Section 2 describes the background concepts of this study, while Section 3 overviews the related work; Section 4 presents the adopted review process methodology. Section 5 presents some bibliographic outcomes whereas Section 6 shows the results of our analysis; Section 7 discusses the key findings, the most interesting challenges and research directions that emerged from this study. Finally, Section 8 concludes the paper.

2. Background

In this section, we briefly describe the analytical techniques encountered during this work. We also describe the kinds of artificial neural networks and the metrics typically used to assess their performance.

2.1. Analytical techniques (features sources)

Several **spectroscopy** techniques operate by shining light of a specific wavelength range onto a sample. The light is absorbed, transmitted, or reflected by the sample, and these interactions can be measured and analyzed to provide information about the sample. Depending on the range used, we distinguish between:

NIR Spectroscopy utilizes the near-infrared region of the electromagnetic spectrum, typically from about 750 nm to 2500 nm (750 nm to 1400 nm for SWIR). It is especially useful for quantifying organic functional groups and analyzing samples with high water content. It allows for rapid, simultaneous determination of several components.

Visible (VIS) Spectroscopy involves the visible region of the electromagnetic spectrum, approximately from 400 to 750 nm. It is commonly

Nomenclature

Neural Networks (NN)	
1D-CNN	1-Dimensional Convolutional NN
AANN	Auto Associative NN
ANFIS	Adaptive Neuro-Fuzzy Inference System
ANN	Artificial NN
BPNN	Back Propagation NN
CFNN	Cascade Forward NN
CNN	(2-dimensional) Convolutional NN
FCNN	Fully Connected NN
FNN	Feedforward NN
LSTM	Long Short-Term Memory NN
MLP	Multi Layer Perceptron
RBM	Restricted Boltzmann Machine
SOM	Self-Organizing (Kohonen) Map
Metrics	
AUC	Area Under the ROC Curve
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MSE	Mean Squared Error
RER	Relative Error Reduction
RMSEP	Root Mean Square Error of Prediction
RPD	Ratio of Performance to Deviation
Analytic methods	
EEMs	Excitation-Emission Matrix fluorescence spectroscopy
FTIR	Fourier Transform Infrared spectroscopy
GC	Gas Chromatography
GC–IMS	GC – Ion Mobility Spectrometry
HPLC	High-Performance Liquid Chromatography
HRMS	High Resolution Mass Spectrometry
LF-NMR	Low field Nuclear Magnetic Resonance
LIFS	Laser-Induced Fluorescence Spectroscopy
NIR	Near-infrared spectroscopy
NMR	Nuclear Magnetic Resonance
SERS	Surface-Enhanced Raman Spectroscopy
SWIR	Short-Wave Infrared spectroscopy
UV	Ultra-Violet spectroscopy
VIS	Visible spectroscopy
VOC	Volatile Organic Compounds
Chemometric methods	
CART	Classification and Regression Tree
DT	Decision Tree
GA	Genetic Algorithm
LDA	Linear Discriminant Analysis
LS	Least-Squares
LS-SVM	Least-Squares SVM
ML	Machine Learning
MLR	Multiple Linear Regression
NB	Naïve Bayes classifier
PCA	Principal Component Analysis
PLS	Partial Least Squares
PLS-DA	PLS Discriminant Analysis
PLSR	Partial LS Regression
RF	Random Forest classifier
SVM	Support Vector Machine
YOLO	'You Only Look Once'
k-NN	k-Nearest Neighbors

Other abbreviations	
EVO	Extra Virgin Olive
EVOO	Extra Virgin Olive Oil
LOO	Lampante Olive Oil
OO	Olive Oil
OSI	Oxidative Stability Index
PDO	Protected Designation of Origin
PV	Peroxide Value
VOO	Virgin Olive Oil

used to determine the concentration of colored compounds in solution and identify metal ions.

UV Spectroscopy utilizes the ultraviolet region of the spectrum, from around 10 to 400 nm. It provides information about conjugated systems, double and triple bonds, and aromatic systems.

In *Fourier Transform Infrared (FTIR) Spectroscopy in the Mid-Infrared (MIR) region with Attenuated Total Reflectance (ATR)* the sample is brought into close contact with a crystal material (typically diamond or germanium) and the light undergoes multiple reflections inside the crystal, each time penetrating a short distance into the sample. The resulting spectrum contains accurate high-resolution absorption features of the sample.

Terahertz spectroscopy falls within the terahertz (THz) region of the electromagnetic spectrum, which lies between the microwave and infrared ranges (approximately 0.1 to 10 THz). It has the ability to penetrate a wide variety of non-conducting materials without the damage associated with ionizing radiation such as X-rays.

Fluorescence spectroscopy is a technique that uses fluorescence to measure a compound concentration in a sample. It involves using ultraviolet light to excite electrons in certain molecules, causing them to emit light at a longer wavelength, usually in the visible spectrum. This non-destructive, rapid method provides insights into the chemical composition of olive oil, including pigments, phenols, and vitamins. In **Laser-Induced Fluorescence Spectroscopy (LIFS)**, a laser is used to induce fluorescence in the sample.

Excitation-Emission Matrix fluorescence spectroscopy (EEMs) also known as **3-D fluorescence spectroscopy**, is a more comprehensive variant of basic fluorescence spectroscopy. EEMs systematically measures fluorescence intensity over a range of excitation and emission wavelengths, creating a two-dimensional matrix or a three-dimensional plot. It can provide detailed information about the composition and interactions of various components within a sample.

Nuclear magnetic resonance (NMR) is a spectroscopic technique in which atomic nuclei in a strong external magnetic field absorb and re-emit electromagnetic radiation at a specific frequency. The absorbed energy perturbs the nuclear spin state, which then returns to equilibrium while emitting a radio-frequency signal. Analysis of this signal can provide detailed information about molecular structure, dynamics, reaction states, and chemical environment. *Proton NMR (^1H NMR) Fingerprinting* and *Carbon-13 NMR (^{13}C NMR) Fingerprinting* are based on magnetic properties of hydrogen (protons) and carbon-13 atoms, respectively. ^{13}C NMR is less sensitive, as only about 1.1% of carbon atoms in a natural sample are carbon-13, but it still provides essential information, since carbon is the backbone of organic molecules.

Low-field magnetic resonance (LF-NMR) is a method of imaging that utilizes magnetic fields with lower intensities than conventional NMR which typically operates at field strengths from 1.5 to 5 T. LF-NMR systems typically function at field strengths less than 0.5 T, often down to 0.1 T or even lower.

E-nose and e-tongue An electronic nose (e-nose) is a device designed to detect and discriminate among complex odors using an array of sensors, generally composed of various types of chemical sensors. Each sensor in the array responds to a specific volatile compound, or to a class of compounds. When an odor sample is analyzed, the sensors generate a pattern or 'fingerprint' that can be used to identify the odor. An e-tongue operates on a similar principle as an e-nose but is designed to analyze liquid samples.

Dielectric spectroscopy is a non-invasive technique that studies a sample's interaction with an applied electric field by measuring its dielectric properties (permittivity and conductivity) as a function of frequency, and sometimes temperature. An alternating electrical field is applied to the sample, and its response—how it stores and dissipates energy—is analyzed. This response reveals information about the sample's molecular dynamics, polarization processes, and conduction mechanisms.

Raman spectroscopy is a non-destructive technique used to study vibrational, rotational, and other low-frequency modes in a system. It provides detailed information about molecular vibrations, helping identify a sample's chemical composition and structure. In this method, laser light is shone on a sample, and the scattered light is collected and directed through a monochromator. The resulting spectrum, plotted as intensity versus frequency shift, provides a unique Raman spectrum that serves as a "fingerprint" for the substance.

Surface-Enhanced Raman Spectroscopy (SERS) is a variation of Raman spectroscopy that increases the intensity of the Raman scattering. It achieves this enhancement by using nanoparticles or roughened metal surfaces. When molecules are adsorbed on these surfaces, typically gold or silver, the Raman signal can be enhanced by several orders of magnitude.

Ultrasound pulse-echo technique is a non-destructive method for assessing liquid characteristics. An ultrasound transducer sends a high-frequency pulse into the sample, which reflects back from any interface it encounters (e.g., the boundary between different layers in a multi-component product). The same transducer detects the reflected pulse. By analyzing the echo's return time and signal intensity, the sample's thickness, density, and other properties can be determined. This technique is widely used in the food industry for quality control, including measuring fat content, detecting contaminants, and determining texture and consistency.

Chemical–physical parameters. To assess the quality, provenience, and possible adulteration of olive oil, various chemical and physical parameters can be used as inputs, such as:

Fatty acid composition: the amounts of different fatty acids, such as oleic acid, linoleic acid, and palmitic acid, are key indicators of quality and can hint at adulteration if the ratios do not match those typical for olive oil.

Phenolic compounds: they contribute to the sensory characteristics of olive oil and also have health benefits. Their composition can help in determining the quality and authenticity of the olive oil.

Peroxide value: the amount of peroxides in the oil can indicate the level of oxidation or rancidity.

Sterol composition: sterols can be used to detect adulteration, to determine the provenance of olive oil, and are related to oxidative stability.

Physical parameters, such as color, turbidity, viscosity, density, and refractive index, that can change with the composition and quality of the oil.

Volatile Organic Compounds (VOC) analysis offers an objective method to complement traditional sensory evaluations and chemical tests in assessing olive oil quality and authenticity.

Gas Chromatography (GC) is used to analyze compounds that can be vaporized without decomposition. In olive oil, GC analyzes the fatty acid profile, volatile compounds, and other flavor, aroma, and quality components. The sample is heated to vaporize it and then carried through a column by an inert gas. Components vaporize at different rates and are separated in the column, with a detector identifying them by their distinct elution times.

Gas Chromatography–Ion Mobility Spectrometry (GC–IMS) adds another layer of separation based on the mobility of ionized molecules in an

electric field, providing a unique spectral fingerprint for each compound. This technique is particularly valuable for detecting trace compounds and complex mixtures.

High-Performance Liquid Chromatography (HPLC) is an analytical technique used to separate, identify, and quantify components in a liquid mixture. It operates under high pressure, allowing for the use of finer particle sizes in the column, leading to improved resolution and faster analyses compared to traditional liquid chromatography.

Image analysis can play a significant role in the assessment of the quality of olives before they are processed into oil. In particular, the size and shape of olives can be indicative of their variety and health, while their color is a critical marker of ripeness, which affects the taste and chemical properties of the oil.

2.2. Neural networks

Artificial neural networks (ANNs), inspired by the biological neural networks of neurons, are designed to mimic the brain's ability to learn from experiences, recognize patterns, and make decisions based on complex data inputs. They constitute an interconnected web of artificial neurons or nodes, where each connection can transmit a signal from one neuron to another. The receiving neuron processes the signal and signals downstream neurons connected to it.

In the landscape of machine learning, ANNs stand apart due to their exceptional ability to learn non-linear and complex relationships. While other machine learning techniques such as DT, SVM, or Linear Regression are powerful in their own right, they often struggle with complex, high-dimensional, and non-linear data. ANNs, on the other hand, excel in these scenarios. Among the various types of neural networks, we briefly outline those we came across in the course of this review.

Feedforward Neural Networks (FNNs) are the simplest type of neural network. Information in FNNs moves in one direction only, from the input layer, through any hidden layers, to the output layer, without any loops. This architecture allows for a broad range of applications, including regression, classification, and pattern recognition tasks.

Multi-Layer Perceptron (MLP) is a specific type of feedforward neural network. It is characterized by having one or more hidden layers of neurons between the input and output layers. Each neuron in a layer is connected to every neuron in the subsequent layer, forming a densely connected structure. MLPs use supervised learning techniques and are typically trained using backpropagation.

Many of the reviewed papers either utilize minor variations of FNN or MLP or refer to them using synonymous terms, such as Back Propagation Neural Network (BPNN), Feedforward Back Propagation Neural Network, or Multi Layer Feedforward Neural Network.

Cascade Forward Neural Networks (CFNNs) are a type of FNN but with a twist: each neuron in the hidden layer is connected to all subsequent neurons and to the output. This creates a ''cascade'' of connections that can often learn faster and achieve higher accuracy than simple FNNs, especially on complex tasks. However, the increased complexity of CFNNs can make them more challenging to train and more prone to overfitting if not handled correctly.

In practice, when most authors refer to MLPs or basic FNNs, they are usually implying a fully connected structure, or a **Fully Connected Neural Network (FCNN)**.

Convolutional Neural Networks (CNNs) on the contrary are usually characterized by a partial connection between layers. CNNs are specially designed to process data with a grid-like topology — an example being an image, which can be understood as a 2D grid of pixels. CNNs are equipped with convolutional layers that apply filters to the input data. These filters can extract features such as edges, corners, and other spatial hierarchies, enabling CNNs to effectively analyze visual data. Here, we assume that CNNs are bidimensional unless specified otherwise.

1D-CNNs, as the name suggests, apply convolution operations to one-dimensional data. Just like their 2D counterparts, they are excellent at identifying local features within the data. However, instead of scanning images, they process data such as time-series signals or text, where spatial relationships exist in one dimension. For instance, they can recognize trends or anomalies in sequential data, making them highly applicable in tasks such as natural language processing, time-series forecasting.

Auto Associative Neural Networks (AANNs), also known as Autoencoders, are a type of neural network used for learning efficient codings of input data. These networks are unsupervised learners, meaning they do not require labeled data. An AANN consists of an encoder, which compresses the input into a latent-space representation, and a decoder, which reconstructs the input data from this latent representation. By learning to reconstruct the input data, AANNs can capture the most important features of the data, making them useful for dimensionality reduction, anomaly detection, and even generative tasks.

Long Short-Term Memory Neural Networks (LSTMs) are a type of recurrent neural network (RNN) architecture that is specifically designed to address the issue of long-term dependencies and vanishing gradient problem found in traditional RNNs. The LSTM architecture is particularly effective in learning from sequences of data and is widely used in applications such as natural language processing, speech recognition, and time series prediction.

In this review, we have also included papers on the following three architectures because they are often categorized under the broad umbrella of neural networks, despite their structural differences from Feedforward Neural Networks (FNN) and Convolutional Neural Networks (CNN).

Restricted Boltzmann Machines (RBMs) are a type of generative stochastic artificial neural network. They are characterized by two layers: a visible layer (inputs) and a hidden layer, with connections only between layers and not within. The ''restriction'' in RBMs eliminates connections between nodes in the same layer, simplifying the learning process. RBMs are commonly used in feature learning, classification, collaborative filtering, and even in the initial stages of training deeper neural networks. Their unique structure and stochastic approach allow them to model complex, high-dimensional data and generate new samples that are similar to the input data.

Self-Organizing Maps (SOMs), also known as Kohonen Maps, are a type of artificial neural network that is trained using unsupervised learning. SOMs are primarily used for data visualization and dimensionality reduction. They are unique in their ability to maintain the topological features of the input space in the output space, providing a kind of ''map'' of the data. Each neuron in the SOM learns to represent different regions of the input space, and neurons that are close together in the network represent regions that are close together in the input space.

Adaptive Neuro-Fuzzy Inference System (ANFIS) is a kind of artificial neural network that is based on Takagi–Sugeno fuzzy inference system. It combines the learning capabilities of neural networks with the uncertainty handling and approximate reasoning of fuzzy logic.

2.3. Performance metrics

The performance metrics used to evaluate the outcomes are strictly dependent on the type of problem the ANN has been developed for. For regression problems, MSE and MAE are the most suitable metrics, while for classification problems the most used metric is accuracy. In the following, we report the performance metrics we encountered in this review.

Mean Squared Error (MSE) measures the average squared difference between predicted and actual values. **RMSE** is the square root of MSE and its formula coincides with that of RMSEC (Root Mean Squared Error of Calibration), where the difference is between predicted values and values used for model calibration.

Mean Absolute Error (MAE) measures the average absolute difference between predicted and actual values.
Root Mean Square Error of Prediction (RMSEP) is a measure of the prediction error for a regression or calibration model, reflecting the difference between observed and predicted values in a validation dataset.
Ratio of Performance to Deviation (RPD) is the ratio between the standard deviation and RMSEP.
Accuracy represents the proportion of correctly classified instances across all classes.
Precision measures the accuracy of positive predictions among the instances predicted as positive.
Sensitivity (Recall or **True Positive Rate)** measures the proportion of actual positive instances correctly predicted.
Specificity (True Negative Rate) measures the proportion of negatives that are correctly identified by the model.
F1 Score is the harmonic mean of precision and sensitivity, providing a balance between the two.

The **Kappa value** or Cohen's Kappa, is a statistic that measures inter-rater agreement for categorical items.
Area Under the ROC Curve (AUC) is a performance measurement for classification problems at various threshold settings. The Receiver Operating Characteristic (ROC) curve plots the true positive rate (sensitivity) against the false positive rate (1 - specificity) at different threshold levels.

A **confusion matrix** is a table often used in classification problems that compares the actual target values with those predicted by the ANN model. From the confusion matrix, several metrics can be calculated, such as accuracy, precision, sensitivity (recall), specificity, F1-score, and more.
Mean Absolute Percentage Error (MAPE) measures the percentage difference between predicted and actual values.
R-squared (R^2) measures the proportion of the variance in the dependent variable that is predictable from the independent variable.
Relative Error Reduction (RER) calculates the proportion by which a model reduces the error compared to a baseline model.

3. Related work

Recent studies have addressed the problems of olive oil quality, safety, and authenticity. There is a rapid growth in the number of research articles that adopt multi-element and multi-analysis techniques, including artificial neural networks, for detecting adulteration, for tracing the geographical origin of the olive oil, and its authentication and classification. Although several surveys exist, no previous work has provided an up-to-date systematic review and classification of ANN techniques for olive oil production and characterization.

Gonzalez-Fern et al. (2019) provides a critical review of the use of artificial intelligence in the olive oil production, classification and authentication. Here, the works found are divided into five categories: (i) application of ANNs in olive oil production; (ii) ANNs for characterizing olive oil and analyzing its composition; (iii) ANNs for authentication, classification and detecting adulteration; (iv) application of ANNs in protected designations of origins; (v) ANNs and sensory devices for fingerprinting in olive oil quality control. In the review paper (Jiménez-Carvelo et al., 2019), data mining and machine learning methods are collated with respect to the analytical evaluation of food quality and authenticity, including olive oil. After explaining the terminology and the main differences between data mining and machine learning (usually referred to collectively as *chemometric techniques*), the focus shifts to some of the most commonly used methods, namely: SVM, CART, and RF. Rocha et al. (2020) is a review on the application of non-linear methods, such as ANN, SVM and SOM, to food analysis. This review focuses one the use of these methods in classifying the geographical origin and determining the safety properties of food products. The authors of Tahir et al. (2022) offer a systematic review of papers published in the period 2013–2020 addressing various analytical tools for verifying the geographical origin of oils. They conclude that PCA, PLS-DA and LDA are the chemometric approaches mostly used for oil authentication, whereas RF, SVM, LS-SVM, ANNs, and Support Vector Classifier, have been scarcely used, although they have been proven able to provide good classification models. Meenu et al. (2019) is a critical review on the analytical techniques used to detect adulteration of extra-virgin olive oil. In this review, different types of vibrational spectroscopy (infrared, Raman, FTIR) were analyzed, as well as other spectroscopic techniques such as ultraviolet–visible, nuclear magnetic resonance, mass and fluorescence spectroscopies. Also chromatographic techniques, such as HPLC, gas chromatography, and their combined application is investigated. Other types of analytical techniques considered are those based on thermal methods such as Differential scanning calorimetry and e-nose. Ordoudi et al. (2023) is a systematic review on the use of FTIR spectroscopy for detecting adulteration in virgin olive oil, with no filter applied to the publication year in the databases searches. A total of 32 journal articles covering a time of 27 years (the first one is dated 1994) have been categorized according to the following five points of view: the field of application, namely the botanical origin or process type of adulterant(s); the sampling methodology and the types of mixtures (binary, ternary, and concentration ranges); the procedure of data acquisition (type of instrument, type of cell, instrument parameters); the pre-processing schemes applied to spectral data; the modeling and validation schemes. Sanaeifar et al. (2017) is a review on the use of the e-nose for detecting food contamination. The categories considered are: meat and fish, milk and dairy products, fresh and processed fruits and vegetables, beverages, grains and bakery products, and vegetable oils.

Several review papers focus on the analytical techniques used to test raw food and/or finished products, such as spectrometries and chromatographies. In Kalogiouri et al. (2020), the focus is on the use of High Resolution Mass Spectrometry (HRMS) in food authenticity assessment, and in particular the olive oil, since it allows the identification of a large number of constituents. In Esteki et al. (2018) studies on the use of chromatography methods for food authentication are reported: they considered edible oils and fats, dairy products, honey, beverages, meat and processed meat. As chemometric tools, they considered both linear ones, such as PCA, LDA, and PLS, and nonlinear ones, such as ANNs. In Avramidou et al. (2018) the focus is instead on molecular methods, such as SSR, AFLP, RAPD, which require complex multivariate analyses. In the first part of the review, the authors comprehensively describe both linear and non-linear methods that could be applied to molecular data, while in the second part the focus is on the molecular markers used for assessing olive oil products. The main objective of the review (Martín-Torres et al., 2022) is to overview studies presenting chemometric techniques and multivariate approaches used to gain insight on the shelf life of both virgin and refined vegetable edible oils under forced and ambient conditions of storage. In the conference paper (Natarajan & Ponnusamy, 2022), a review about the last two years of works on the adulteration of vegetable oils is presented; it considers spectroscopic, imaging, and DNA-based methods and the machine learning algorithms for determining oil adulteration. In Stilo et al. (2021), the review focuses on the application of chromatographic fingerprinting of 2D peak patterns to determine the composition details of food samples, including EVO oil.

The work most similar to ours is Gonzalez-Fern et al. (2019). However, this review, dated 2019, is not current and does not employ a systematic methodology. For a better comparison, we summarize related works in Table 1. As shown in the last row, this paper provides an up-to-date and systematic analysis of the literature on the application of ANNs in the specific domain of olive oil.

Table 1
Comparison with related works.

Ref.	Year	Aim of review	Review approach	Focus on				
				OO	Food analysis	ANN techniques	Other ML techniques	Analytical techniques
Gonzalez-Fern et al. (2019)	2019	ANN techniques for olive oil production, classification and authentication	Critical review	✓	–	✓		
Jiménez-Carvelo, González-Casado, Bagur-González, and Cuadros-Rodríguez (2019)	2019	Data mining/machine learning techniques for food quality and authentication	Review	–	✓	–	✓	–
Rocha, do Prado, and Blonder (2020)	2020	Non-linear methods for geographical classification	Review	–	✓	✓		
Tahir et al. (2022)	2022	Analytical tools and chemometrics for geographical classification	Systematic search	✓	–	✓		✓
Meenu et al. (2019)	2019	Analytical techniques to detect adulteration	Critical review	✓	–	–	–	✓
Ordoudi et al. (2023)	2023	FTIR spectroscopy for detecting adulteration	Systematic search	✓	–	–	–	✓
Sanaeifar et al. (2017)	2017	E-nose for detecting food contamination	Review	–	✓	–	–	✓
Kalogiouri, Aalizadeh, Dasenaki, and Thomaidis (2020)	2020	HRMS for assessing EVOO authenticity	Review	✓	–	–	✓	✓
Esteki et al. (2018)	2018	Chromatographic methods for food authentication	Review	–	✓	✓	✓	✓
Avramidou, Doulis, and Petrakis (2018)	2018	Chemometric and molecular methods for detecting adulteration in olive-based products	Review	✓	–	–	✓	✓
Martín-Torres et al. (2022)	2022	Studies on shelf life of edible oils	Review	–	✓	–	✓	✓
Natarajan and Ponnusamy (2022)	2022	ML techniques for detecting adulteration in edible oils	Review	–	✓	✓	✓	–
Stilo et al. (2021)	2021	Chromatographic methods for food composition	Review	–	✓	–	✓	✓
This	*2024*	*ANN techniques for olive oil processing and post-production analyses*	*Systematic search + snowballing*	✓	–	✓	–	✓

4. Review process

In conducting our review, we followed the well-known guidelines by Kitchenham and co-authors (Kitchenham, 2004) for performing systematic literature reviews. First, we defined the research goal and the research questions, then we formulated the search query and selected the primary studies, finally we extracted and analyzed the results.

Our research goal is to characterize the application of ANNs in the olive oil processing. More precisely, we are interested in understanding whether (and how) ANNs are applied during the olive oil processing, which are the main objectives, what are the adopted types of ANN and their inputs, what are the adopted performance metrics and in which phases of the olive oil processing ANNs are applied.

To reach our research goal, we formulated the following Research Questions (RQs):

RQ1: What are the phases of the olive oil processing in which ANNs are mostly adopted?

RQ2: What are the main goals of using ANNs in the olive oil processing and the used input types?

RQ3: What are the main types of ANNs that are used during the olive oil processing?

RQ4: What metrics are adopted to evaluate the ANN model performance?

According to these research questions we formulated the search query presented in Table 2 (left part). We executed this search query on six electronic databases, that are: Scopus (http://www.scopus.com), Web of Science (https://www.webofscience.com), IEEE Xplore (http://ieeexplore.ieee.org), ACM (http://dl.acm.org), ScienceDirect (https://www.sciencedirect.com/), PubMed (https://pubmed.ncbi.nlm.nih.gov/), searching for title, abstract and keywords. In each of these digital libraries, we searched for English-language papers published from January 2017 to December 2023, using our search query. Some digital libraries impose limitations when performing queries. For instance, SpringerLink does not allow the use of parentheses, ScienceDirect accepts a maximum of eight logical connectors in search queries, and the ACM Digital Library does not accept wildcards. We tailored our search strategy to accommodate the unique constraints of each digital library, obtaining a total of 570 studies, as illustrated in Fig. 1. We exported the bibliographic entries to an Excel spreadsheet for analysis and manually removed 264 duplicate studies.

During the first step of our selection process, each author reviewed the title, abstract, and keywords of randomly chosen subset of papers, applying the inclusion and exclusion criteria described in Table 2 (right part). To ensure consistent application of these criteria, plenary meetings were held among all authors to discuss their findings, reach consensus, and address any uncertainties. In this first step, a total of 157 studies were selected for a full-text review.

In the second step, the authors independently applied the same criteria of Table 2 (right part) to the 157 studies obtained in the previous iteration by thoroughly reading them. At the end of this second round, 66 studies were selected for inclusion.

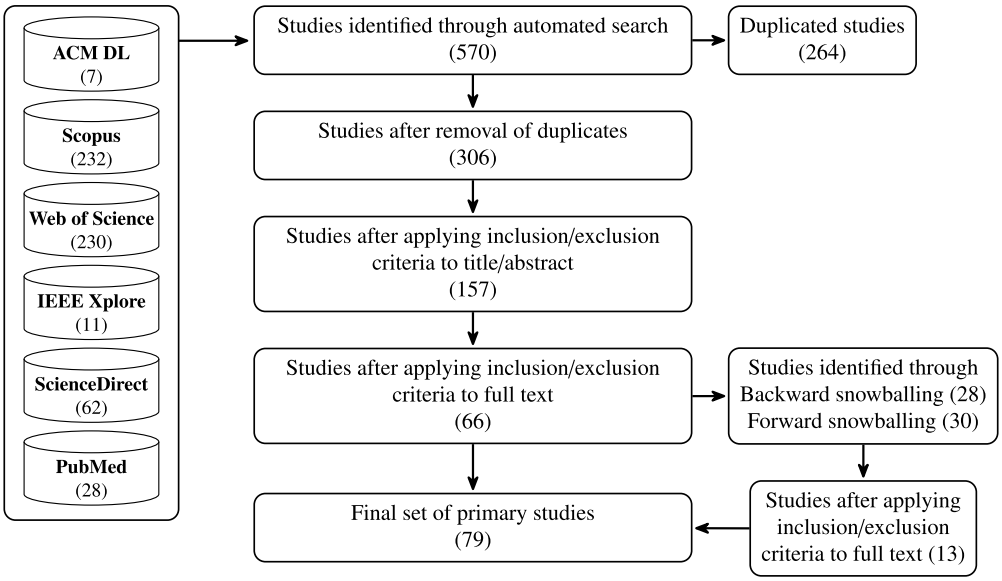


Fig. 1. Selection process and numerical outcomes.

Table 2
Search query and inclusion/exclusion criteria.

Search query	Inclusion/Exclusion Criteria
(evoo < *OR* > "olive oil*" < *OR* > "EVO oil*")	**Inclusion Criteria**
< *AND* > ("deep learning" < *OR* > "machine learning"	Studies on techniques/algorithms/frameworks that apply ANNs to the olives and olive oil
< *OR* > "artificial intelligence" < *OR* > AI	Studies on olive oil production, characterization and authentication leveraging ANNs
< *OR* > "neural networ*" < *OR* > "reinforcement learning"	**Exclusion Criteria**
< *OR* > mlp < *OR* > ANN < *OR* > "*supervised learning"	Studies not concerned with olives and olive oil
< *OR* > "inference system*" < *OR* > "self-organizing map*"	Studies adopting machine learning techniques different from ANNs
< *OR* > SOM < *OR* > kohonen < *OR* > AANN	
< *OR* > "learning algorithm*")	Studies addressing olive tree cultivation and diseases
	Not primary studies (surveys, editorials, panels, theses, white papers, books, etc.) or not peer-reviewed studies

Finally, to complement the results of the automated search, we performed a round of backward and forward snowballing on the 66 papers identified in the second selection step, according to the guidelines for snowballing in systematic literature reviews presented in Wohlin (2014). Specifically, each author performed snowballing on a balanced and randomly chosen subset of papers found in the second step, analyzing for backward snowballing its list of references, and for forward snowballing its citations in Google Scholar (http://scholar.google.com). This approach enabled us to identify relevant studies that might have been published in venues not indexed by our selected databases or were indexed under keywords not included in our initial search query. After removing duplicates, we obtained 58 studies (28 and 30 studies from backward and forward snowballing, respectively). We examined the text of these 58 papers applying the inclusion/exclusion criteria of Table 2 (right part) and we found 13 additional studies. These were added to the initial set of 66 studies, resulting in a total of 79 primary studies.

Fig. 1 summarizes the numerical outcomes of the selection process. We analyzed the set of primary studies by extracting this information: a unique identifier of the paper, the title, the year of publication, the publication venue, the authors, the main objective and contribution. The results of our analyses are presented in Sections 5 and 6.

5. Bibliometric analysis

In this section, we report the results of several quantitative analyses we performed on the set of the primary studies.

The majority of primary studies (71) have been published in 41 different journals, while 8 primary studies have been published in 6 different conferences. Table 3 shows, in alphabetical order, the journals in which the primary studies have been published. Among the journals, 21 are ranked as *Q1*, 12 as *Q2*, 7 as *Q3* and 1 as *Q4*, according to the most recent Scimago Journal & Country Ranking (https://www.scimagojr.com/). *Food Chemistry (Elsevier)* has the highest number of primary studies (9). Among the conferences (not reported in Table 3), *SPIE Photonics Europe* featured the highest number of primary studies (3 studies), while each of the other five conferences (Congreso Argentino de AgroInformática, IEEE Int. Conf. on Awareness Science and Technology, IEEE Int. Conf. on Omni-layer Intelligent Systems, Int. Conf. on Information and Communication Systems, Int. Work-Conference on the Interplay Between Natural and Artificial Computation) published only one primary study.

Fig. 2 (on the left side) shows the trend of publications on ANNs applied to olive oil over the past 7 years (considering their first online appearance). We observe that the number of publications increased along the years with two peaks in 2019 (14 primary studies) and 2022 (17 primary studies). Fig. 2 (on the right side) shows the number of primary studies according to the score of their publication source, as well as the number of primary studies presented at conferences.

The 79 primary studies were written by 266 unique authors. Fig. 3 displays the distribution of the authors' affiliation countries, indicating a widespread global interest in applying ANNs to olive oil research. The most active contributors are from China (about 23% of the total number of authors), Italy (about 19%), and Spain (about 16%), followed by Iran (about 9%) and Turkey (about 5%). We can notice that the most active

Table 3
Publication sources (journals).

Source title	Score	#	Source Title	Score	#
Agricultural Engineering International:CIGR Journal - PKP Publishing Services Network	Q3	1	J. of Chemometrics - Wiley	Q3	2
AgriEngineering - MDPI	Q2	1	J. of Food Composition and Analysis - Elsevier	Q1	2
Analytica Chimica Acta - Elsevier	Q1	1	J. of Food Engineering - Elsevier	Q1	1
Analytical Methods - Royal Society of Chemistry	Q2	1	J. of Food Measurement and Characterization - Springer	Q2	3
Biosystems engineering - Elsevier	Q1	1	J. of Food Process Engineering - Wiley	Q2	1
Chemical Papers - Springer	Q2	1	J. of Food Quality - Hindawi	Q2	1
Chemosensors - MDPI	Q2	1	J. of food science and technology - Wiley	Q1	3
Computers and Electronics in Agriculture - Elsevier	Q1	3	J. of Near Infrared Spectroscopy - SAGE Publications Inc.	Q3	1
Earth Science Informatics - Springer	Q2	1	J. of the Science of Food and Agriculture - Wiley	Q1	2
Engineering in Agriculture, Environment and Food - J-STAGE	Q3	1	LWT - Food Science and Technology - Elsevier	Q1	3
European Food Research and Technology - Springer	Q2	4	Microchemical Journal - Elsevier	Q1	1
Food Analytical Methods - Springer	Q2	2	Neural Computing and Applications - Springer	Q1	1
Food Chemistry - Elsevier	Q1	9	Postharvest Biology and Technology - Elsevier	Q1	2
Food Control - Elsevier	Q1	2	Quality Assurance and Safety of Crops & FoodsCodon Publications	Q2	2
Food Science & Nutrition - Wiley	Q1	1	Química Nova - PubliSBQ	Q3	1
Foods - MDPI	Q1	4	Rivista Italiana Delle Sostanze Grasse - Innovhub SSI	Q4	1
Frontiers in chemistry - Frontiers Media S.A.	Q1	1			
IEEE SENSORS JOURNAL	Q1	1	Scientia Horticulturae - Elsevier	Q1	1
Information Processing in Agriculture - Elsevier	Q1	1	Sensors and Actuators B: Chemical - Elsevier	Q1	1
Int. J. of Advanced Computer Science and Applications - SAI	Q3	1	Spectrochimica Acta Part A: Molecularand Biomolecular Spectroscopy - Elsevier	Q2	2
Int. J. of Postharvest Technology and Innovation - Inderscience	Q3	1	Talanta - Elsevier	Q1	1

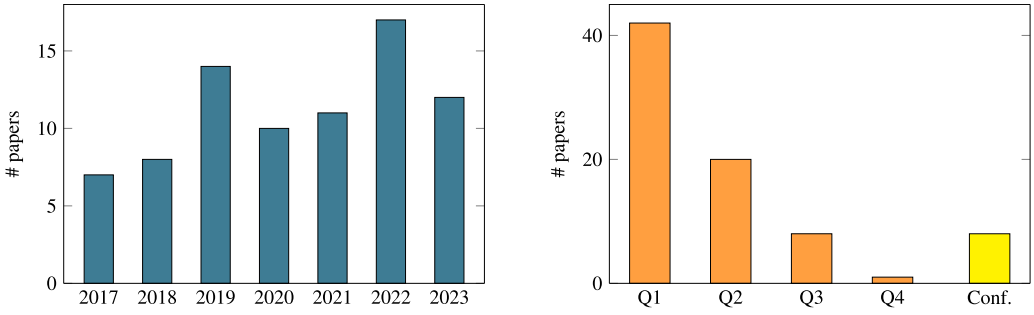


Fig. 2. Distributions of primary studies first online appearance (on the left), and of publications sources scores (on the right).

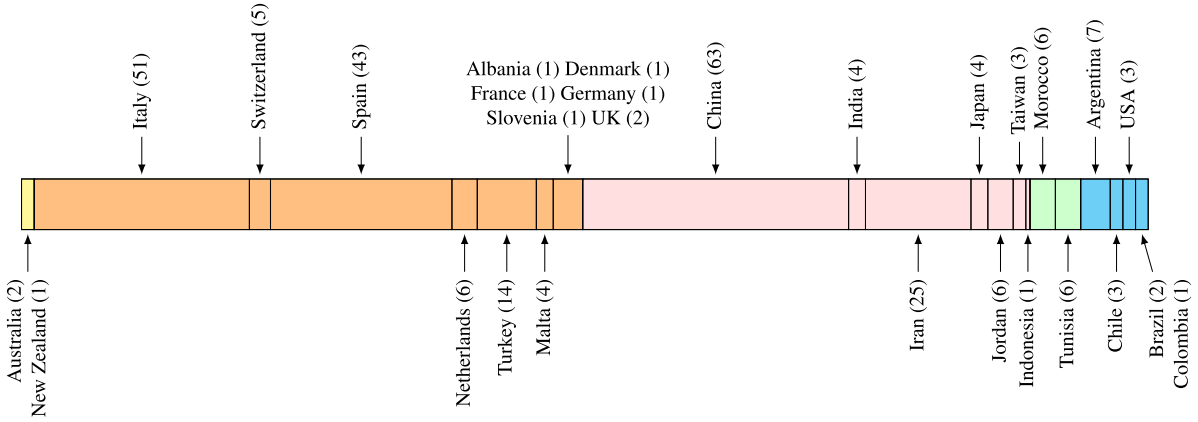


Fig. 3. Distribution of the affiliation country of the authors of the primary studies (a few have double affiliation), not counting multiple papers authored.

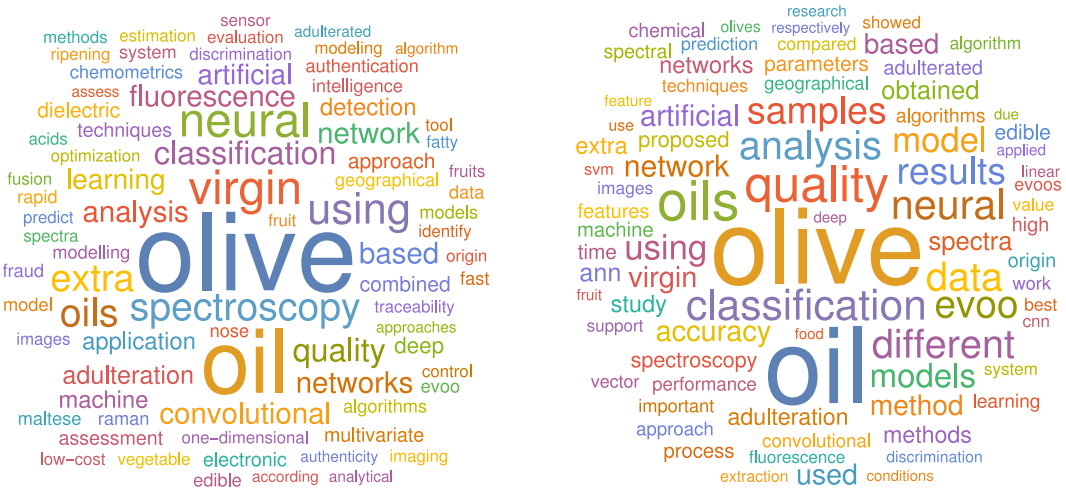


Fig. 4. Word clouds of the 80 most common words in titles (left) and abstracts (right).

contributors are from the world's areas that are also among the most active in olive oil production. Spain and Italy are among the world's leading producers of olive oil whereas in China olive oil yields grew in the last years. It is noteworthy that Greece is absent from this figure, despite being one of the main European producers of EVOO.

Finally, to complement the above bibliometric analyses, Fig. 4 presents word clouds that visually summarize the 80 most frequent terms or keywords found in the titles and abstracts of the analyzed primary studies. We can observe that apart from the "olive" and "oil" terms, the larger words highlight prevalent themes or topics: for instance fluorescence spectroscopy among analytical techniques, classification and adulteration as key objectives, and accuracy as the main performance metric. These findings align with the deeper analysis that will be presented in the next section.

6. Results

In this section, we describe the qualitative results of our review and we answer the research questions presented in Section 4.

In Tables 4 and 5, we have organized the final set of primary studies (column labeled Ref#) according to four aspects: phase (column labeled Phase), objective (column labeled Objective), main input type (column labeled Input type), and the neural network used (column labeled ANN).

We have divided the papers into two subsets: papers dealing with olive *oil itself* (see Table 4), and papers concerned with aspects of *oil production* (see Table 5).

While a few papers explore various avenues, most of them focus on a single goal, which we have classified as follows:

- Adulteration detection (**ADU**): The main goal is to distinguish between pure extra-virgin olive oil (EVOO) or virgin olive oil (VOO) and mixtures containing only a fraction of olive oil.
- Classification (**CLASS**): Studies where the goal is to discriminate between olive oil and other types of oil, or between different grades of olive oil (EVOO vs. VOO vs. LOO).
- Cultivar discrimination (**CULT**): Papers that aim to identify the olive cultivar either from oil characteristics or from olive fruit images.
- Geographical classification (**GEO**): Studies that aim to classify a set of olive oils based on the geographical origin of the processed olives.
- Attributes prediction (**ATTR**): The objective is to determine characteristics tied to the oil or olive quality, such as the amounts of polyphenols or peroxides, or the ripeness stage.
- Oxidative stability (**STAB**): The purpose is to determine the oxidative stability of the oil, which is also connected to its shelf life.
- Optimization (**OPT**): The goal is to optimize specific aspects of oil production.

The principal input types (features sources), neural network architectures, and performance metrics used in the primary studies are summarized in Sections 2.1, 2.2, and 2.3, respectively.

We have decided to adhere to the nomenclature used by authors in their papers, even when certain terms referring to a specific neural network model likely denote the same architecture. This is observed in instances such as FNN, MLP, and BPNN. In Tables 4 and 5, entries labeled 'ANN' mean that the authors did not specify further. ANN name followed by an asterisk, as in CNN*, signals cases in which the authors propose a new variant of a known ANN model.

In the following, we answer to the research questions presented in Section 4.

6.1. RQ1 - What are the phases of the olive oil processing in which ANNs are mostly adopted?

The phase of olive oil processing where ANNs are most frequently adopted is the post-production phase, meaning that ANNs are mainly applied to olive oil samples. The majority of the primary studies (approximately 82%) focus on olive oil (Table 4). Here, ANNs are mainly used for identifying adulterated olive oil, discriminating between olive oil and other oil types, or classifying olive oils based on their geographical origin or cultivar.

In 12 primary studies (about 15% of the total number), ANNs are applied during the olive oil production phase (Table 5), for instance for classifying olive fruits according to their ripeness or for the optimization of the clarification step in the olive oil production. Finally, two primary studies (Martínez Gila et al., 2020; Martínez Gila et al., 2021) adopt ANNs for both the classification of olive fruits and oils based on their fatty acid ethyl ester content, and determining whether they are within the legal limits.

6.2. RQ2 - What are the main goals of using ANNs in the olive oil processing and the used input types?

Regarding the post-production analysis of olive oil, which is the focus of the majority of the papers (67 in total), the most commonly used techniques for obtaining features for the ANNs are: NIR/VIS/UV/SWIR spectroscopy (14 papers), fluorescence spectroscopy (13 papers), chemical/physical analyses (15 papers), and NMR/LF-NMR (6 papers). It is

Table 4
Classification of primary studies concerning oil characterization.

Ref#	Objective	Input type	ANN	Ref#	Objective	Input type	ANN
Kumar (2017)	ADU	HPLC	SOM	Cervera-Gascó, Rabadán, López-Mata, Álvarez-Ortí, and Pardo (2023)	CULT	CHEM	MLP
Lia, Morote Castellano, Zammit-Mangion, and Farrugia (2018)	ADU	EEMs	ANN	Senizza, Ganugi, Trevisan, and Lucini (2023)	CULT	CHEM	FNN
Xu, Hassan, Kutsanedzie, Li, and Chen (2018)	ADU	FTIR	BPNN	Acquarelli et al. (2017)	GEO	FTIR	1D-CNN*
Capar, Kavuncuoglu, Yalcin, and Toga (2019)	ADU	CHEM	MLP	Gumus, Ertas, Yasar, and Gumus (2018)	GEO	CHEM	ANN
Soltani, Rashvand, Teimouri, and Omid (2019)	ADU	dielectric s.	MLP	Liu et al. (2018)	GEO	THz s.	BPNN
Aroca-Santos, Lastra-Mejías, Cancilla, and Torrecilla (2019)	ADU	VIS	MLP	Torreblanca-Zanca et al. (2019)	GEO, ADU	fluorescence s., VIS	MLP
Izquierdo et al. (2020)	ADU	thermography	CNN	Sayago, González-Domínguez, Urbano, and Fernández-Recamales (2019)	GEO, CULT	CHEM, NMR	MLP
Zarezadeh, Aboonajmi and Ghasemi Varnamkhasti (2021)	ADU	ultrasound	ANN	Gumus, Yasar, Gumus, and Ertas (2020)	GEO	CHEM	MLP
Liu, Yao, Xia, Gao, and Gong (2021)	ADU	NIR	CNN	Harrington (2020)	GEO	CHEM	RBM*
Hou et al. (2021)	ADU	LF-NMR	1D-CNN	Palagano, Valli, Cevoli, Bendini, and Toschi (2020)	GEO	e-nose	MLP
Zarezadeh, Aboonajmi, Varnamkhasti and Azarikia (2021)	ADU	e-nose, ultrasound	MLP	Wang et al. (2020)	GEO	FTIR	1D-CNN*
Violino et al. (2021)	ADU	VIS–NIR, VOC	FNN	Lia, Vella, Zammit Mangion, and Farrugia (2020)	GEO	NMR	FNN
Bl et al. (2022)	ADU	e-tongue	ANN	Violino et al. (2020)	GEO	VIS–NIR	FNN
Omwange et al. (2022)	ADU	EEMs, images	CNN	Lia, Zammit Mangion, and Farrugia (2021)	GEO	FTIR	FNN
Yakar and Karadağ (2022)	ADU	CHEM	ANN	Lia, Zammit-Mangion, and Farrugia (2022)	GEO	fluorescence s.	ANN
Surya and Senthilselvi (2022)	ADU	LF-NMR	LSTM	Yousuff and Babu (2023)	GEO	FTIR	FCNN*
Chuma and Rasmussen (2022)	ADU	NIR, dielectric s.	ANN	Ortenzi, Figorilli, Violino, Pallottino, and Costa (2023)	GEO	VIS–NIR	ANN
Meng et al. (2023)	ADU	FTIR VIS–NIR, EEMs	BPNN	Rashvand, Javanmard, Akbarnia, and Sarami (2019)	ATTR	dielectric s.	MLP
Zarezadeh, Aboonajmi, and Ghasemi-Varnamkhasti (2023)	ADU	e-nose, ultrasound	MLP	Sakar et al. (2022)	ATTR	CHEM	ANN
Wu et al. (2023)	ADU	3-D fluorescence s.	CNN	Sperti, Gucciardi, Michelucci, Venturini, and Deriu (2022)	ATTR	fluorescence s.	FNN 1D-CNN
Wu, Du et al. (2024)	ADU	SERS	LSTM	Venturini, Michelucci, Sperti, Gucciardi, and Deriu (2022)	ATTR	fluorescence s.	1D-CNN
Wu, Zhang et al. (2024)	ADU	Raman s.	LSTM	Arnaud, Michelucci, Venturini, Sperti, and Deriu (2022)	ATTR	fluorescence s.	FNN 1D-CNN 1D-CNN*
Zeng, quan Liu, guang Fan, and Wang (2023)	ADU	Raman s.	1D-CNN*	Gonzalez Viejo and Fuentes (2022)	ATTR	NIR e-nose	ANN
Ordukaya and Karlik (2017)	CLASS	e-nose	MLP	Violino et al. (2022)	ATTR	VIS–NIR	FNN
Chien, Shih, and Tseng (2019)	CLASS	VIS–NIR, SWIR	MLP	Venturini et al. (2023)	ATTR	fluorescence s.	1D-CNN
Vega-Márquez et al. (2019)	CLASS	GC–IMS	CNN	Sanaeifar, Jafari, and Golmakani (2018)	STAB	dielectric s., images	MLP
Vega-Márquez, Nepomuceno-Chamorro, Jurado-Campos, and Rubio-Escudero (2020)	CLASS	GC–IMS	FNN	Aroca-Santos, Lastra-Mejías, Cancilla, and Torrecilla (2018)	STAB	VIS	MLP
Hou et al. (2020)	CLASS	LF-NMR	1D-CNN CNN	Lastra-Mejias et al. (2019)	STAB, ADU	fluorescence s.	MLP
Venturini et al. (2021)	CLASS	fluorescence s.	FNN	Arabameri et al. (2019)	STAB	CHEM	ANFIS
Chen et al. (2022)	CLASS	LIFS	1D-CNN*	Ennouri et al. (2019)	STAB	CHEM	MLP
Pradana-Lopez, Perez-Calabuig, Cancilla, Garcia-Rodriguez, and Torrecilla (2022)	CLASS, ADU	images	CNN	Şahin, Sevgen, and Samli (2019)	STAB	CHEM	MLP
Rashvand, Altieri, Matera, Genovese, and Di Renzo (2023)	CLASS	CHEM, dielectric s.	MLP	Sanaeifar and Jafari (2019)	STAB	dielectric s.	ANN
Binetti et al. (2017)	CULT	NMR, NIR, CHEM	MLP				

noteworthy that some papers utilize a combination of these techniques. Table 6 presents a classification of the main input types by goals in this phase (post-production), and the corresponding classification for pre-production and production is reported in Table 7.

Table 5
Classification of primary studies centered on production phase. The first two studies (Martínez Gila, Gámez García, Bellincontro, Mencarelli, & Gómez Ortega, 2020; Martínez Gila et al., 2021) are also concerned with oil characterization.

Ref#	Objective	Input type	ANN	Ref#	Objective	Input type	ANN
Martínez Gila et al. (2020)	ATTR	e-nose	MLP	Aguilera Puerto, Cáceres Moreno, Martínez Gila, Gómez Ortega, and Gámez García (2019)	ATTR	images	MLP
Martínez Gila et al. (2021)	ATTR	e-nose	MLP	Alkhasawneh (2021)	ATTR	images	CFNN
Funes, Allouche, Beltrán, Aguliera, and Jiménez (2017)	OPT	NIR	FNN	Ghandour et al. (2021)	ATTR	images	CNN
Asgari, Sahari, and Barzegar (2017)	OPT	prod. parameters	MLP	Khosravi, Saedi, and Rezaei (2021)	ATTR	images	CNN*
Funes, Allouche, Beltrán, Aguliera, and Jiménez (2018)	OPT	prod. parameters	MLP	Salvucci et al. (2022)	ATTR	images	CNN
Mora, Aliaga, and Fredes (2017)	ATTR	images	MLP	Figorilli et al. (2022)	ATTR	images	CNN
Pariente, Cancilla, Wierzchos, and Torrecilla (2018)	ATTR	images	MLP	Martínez Gila, Navarro Soto, Satorres Martínez, Gómez Ortega, and Gámez García (2022)	ATTR	images	MLP

Table 6
Classification of input types by objectives (on the left) and by neural network used (on the right), in the post-production phase. MPL row also comprises uses of BPNN, FNN, and FCNN.

Objective	*input*	fluor.s., LIF, EEMs	chem., VOC	NIR/VIS/SWIR/UV s.	e-nose, e-tongue	NMR, LF-NMR	FTIR	dielectric s.	GC–IMS, HPLC	images	ultrasound	Raman, SERS	THz s.	thermography
ADU		6	3	6	3	2	2	2	1	2	3	3	·	1
GEO		2	4	3	1	2	4	·	·	·	·	·	1	·
ATTR		4	1	2	3	·	·	1	·	·	·	·	·	·
CLASS		2	1	1	1	1	·	1	2	1	·	·	·	·
STAB		1	3	1	·	·	·	2	·	1	·	·	·	·
CULT		·	4	1	·	2	·	·	·	·	·	·	·	·

NN archit.	*input*	fluor.s., LIF, EEMs	chem., VOC	NIR/VIS/SWIR/UV s.	e-nose, e-tongue	NMR, LF-NMR	FTIR	dielectric s.	GC–IMS, HPLC	images	ultrasound	Raman, SERS	THz s.	thermography
MLP		6	10	9	5	3	4	4	1	1	·	2	1	·
ANN		2	3	3	2	·	·	2	·	·	·	1	·	·
CNN		2	·	1	·	1	·	·	1	2	·	·	·	1
1D-CNN		5	·	·	·	2	2	·	·	·	1	·	·	·
LSTM		·	·	·	·	1	·	·	·	·	2	·	·	·
ANFIS		·	1	·	·	·	·	·	·	·	·	·	·	·
RBM	·	1	·	·	·	·	·	·	·	·	·	·	·	
SOM	·	·	·	·	·	·	·	1	·	·	·	·	·	

Table 7
Classification of input types by neural network and objectives used in the pre-production and production phases.

Objective	Features sources				NN architecture	Features sources			
	images	e-nose	prod. param.	NIR s.		images	e-nose	prod. param.	NIR s.
ATTR	9	2	·	·	MLP, FNN	4	2	2	1
OPT	·	·	2	1	CNN	4	·	·	·
					CFNN	1	·	·	·

We also categorized the primary studies according to the use of the analytical techniques together with the ANN used in the work, as reported in Tables 6 and 7.

In the following, we describe the works found for each goal.

6.2.1. Adulteration (ADU)

Adulteration is mainly detected by using different kinds of input applied to the oil: infrared spectroscopies (near-IR, FTIR, visible near-IR), fluorescence spectroscopies, magnetic resonance, thermography, and also e-nose. As output, most of them give an answer such as "adulterated" or "not adulterated"; other works give the percentage of adulteration. The ANN architecture mostly used is the FNN, but few works make use of CNN or SOM. Fluorescence spectroscopy has been used to detect adulteration in Lia et al. (2018), Omwange et al. (2022) and Torreblanca-Zanca et al. (2019). In Lia et al. (2018), the authors adulterated 12 Maltese pure EVO oils with various seed oils (sunflower, peanut, soybean, linseed, corn, and rapeseed) using different ratios. The authors then compared the results obtained by applying three different validation methods to the ANNs with those from PCA and PLSR statistical models, finding that in general the best results were obtained with PLSR. In Omwange et al. (2022), 16 pure EVO oils and one VOO used as an adulterant were considered. The samples ranged from pure EVOO to pure VOO at intervals of a 10% adulteration rate. They applied three different types of CNN: a 4-layer CNN, AlexNet, and EfficientNet-b0, considering both images and EEMs to predict the ΔK value and Free Fatty Acids (FFA). The CNN models were compared with PLSR, SVM, and DT models. For ΔK prediction, the highest accuracy was obtained with PLSR, while for FFA, the best results were achieved with SVM or CNN models. In Torreblanca-Zanca et al. (2019), both fluorescence and VIS spectra were used as data inputs. The authors prepared 254 samples of fresh EVO oils adulterated with old olive oils aiming to distinguish the PDO and determine the amount of fresh EVOO. The developed MLP model resulted in an accuracy of 100% in identifying the PDO, with a small error in detecting the amount of fresh oil (average mean percentage error 1.68% and MAE 1.52%).

Dielectric spectroscopy has been used in Soltani et al. (2019) and Chuma and Rasmussen (2022). In Soltani et al. (2019), the EVO oils were adulterated with sunflower and canola oils, with concentration ratios up to 50%. From the dielectric spectroscopy, 43 features were selected, and then several data mining methods were applied: MLP, SVM, and DT. The MLP with one hidden layer showed a classification success rate of 100%, while the other methods had an acceptable but inferior performance compared to the MLP. In Chuma and Rasmussen (2022), a relatively small set of samples was used to assess the efficacy of the developed sensor: 10 samples were pure EVO oils, while the other 15 samples were pure soy, corn, canola, sunflower oils, and mixtures of them with EVOO. They compared ANN, Naïve Bayes Gaussian, k-NN, and SVM, showing that the proposed sensor, combined with microwave dielectric properties, achieves 100% accuracy even with a small number of training samples.

Low Field Nuclear Magnetic Resonance (LF-NMR) has been used in Surya and Senthilselvi (2022) and Hou et al. (2021). In Surya and Senthilselvi (2022), 1777 samples of olive oils, coconut oils, rice oils, sesame oils, and sunflower oils were considered. For each sample, they tested if the related oil was pure or adulterated. The ANN architecture used was an LSTM-based neural network with seagull optimization. They also compared their performance with other studies, showing an improvement of about 8%, reaching an accuracy of 98.6%. In Hou et al. (2021), the authors took 252 oil samples: 72 of pure EVO oils, 90 of EVO oil adulterated with hazelnut oil, and 90 of EVO oil adulterated with sunflower oil. The concentration ratios of adulterants ranged from 10% to 100%. The ANN model used was a CNN, finding that an accuracy of 89.29% is achievable with the proposed methodology.

Several types of spectroscopies have been used as input data in Aroca-Santos et al. (2019), Liu et al. (2021), Meng et al. (2023), Violino et al. (2021) and Xu et al. (2018). Liu et al. (2021) used 2D correlation NIR spectra as input data and two datasets: one with 900 samples of sesame oil from three different geographical areas (300 samples for each area), and another dataset with 600 samples of olive oil, pure (150 samples) or adulterated with soybean, corn, and colza (150 samples for each adulterant) at a ratio varying from 10% to 20%. Three different CNNs were developed, with different numbers and types of middle layers. The results show that geographical classification for sesame oil samples reached an accuracy of 97.3%, while for olive oil adulteration, the performance was 88.5%. The aim of Meng et al. (2023) was to evaluate the performance of several spectroscopic techniques for detecting olive oil adulteration. Specifically, they conducted a comparative study considering FTIR, VIS–NIR, and EEMs, in the context of detecting adulteration in EVO oil with soybean oil as adulterant. They also considered three chemometric methods: PCA, PLS, and ANN. They found that FTIR and VIS–NIR guaranteed a performance of 100% in detecting adulteration. In Aroca-Santos et al. (2019), adulteration was studied by analyzing mixtures of EVO oil with soybean or almond oil, or both. The VIS spectroscopies were used as input data. They showed that MLP is superior to PLS models in detecting adulteration in ternary mixtures. In Violino et al. (2021), 96 samples, including pure EVOOs and oils adulterated with 7 different seed oils at different ratios, were analyzed using VIS–NIR spectroscopy and VOCs (Volatile Organic Compounds) analyses as input. The developed ANN model includes two different FNN with one hidden layer, one for VIS–NIR data and one for VOC data. Both FNNs successfully predicted the adulterated oils and also highlighted the most important features of pure EVOOs, namely the top most significant frequencies (in the range 308–380 nm and 440–480 nm) for VIS–NIR spectra, and the most significant compounds (C5, C6, acetaldehyde, acetic acid) for VOC data. Xu et al. (2018) employed FTIR spectroscopy as data input. They considered 160 oil samples: 40 pure EVOOs and 120 EVOOs adulterated with peanut or rapeseed oils, at different concentration ratios. They developed three models: LDA, BPNN, and LS-SVM, and the best results were achieved with LS-SVM, with a discrimination rate in the prediction set equal to 92.5%, while the 4-8-1 BPNN achieved only 82.5%.

The three papers (Zarezadeh, Aboonajmi, Ghasemi Varnamkhasti, 2021; Zarezadeh et al., 2023; Zarezadeh, Aboonajmi, Varnamkhasti et al., 2021) investigated adulteration detection using different input data. The sample set comprised mixtures of pure EVO oils with sunflower, canola, and corn oils at various concentration ratios (0, 5, 10, 20, 35, and 50%) for a total of 294 samples. In Zarezadeh, Aboonajmi, Ghasemi Varnamkhasti (2021), ultrasound was used as the data input. Seven models were applied: Naïve Bayes, SVM, Gradient Boosting Classifier, k-NN, ANN, logistic regression, and AdaBoost. The highest accuracy in predicting adulteration was achieved with Naïve Bayes, at 90.2%, while the ANN's accuracy was 86.27%. In Zarezadeh, Aboonajmi, Varnamkhasti et al. (2021), e-nose and ultrasound served as input data. Testing the same seven models, the best performance was obtained with gradient boosting, showing an accuracy of 97.75%. Zarezadeh et al. (2023) refined the previous work by also testing linear models, all applied after performing PCA to reduce the number of features. They identified the "percentage of ultrasonic amplitude loss" as having the highest influence on classification. Furthermore, data from the ultrasound system proved more effective than the e-nose. In this instance, the ANN emerged as the most effective classifier, achieving the highest accuracy (95.51%).

Yakar and Karadağ (2022) employed gas chromatography as the input data. They adulterated olive oil with cottonseed, canola, and soybean oils at different concentration ratios (1, 5, 10, 20, and 40%). Three ML models (SVM, k-NN, and DT) were developed for classification, and an ANN was utilized for regression.

Capar et al. (2019) used rheology parameters as data input, such as the refractive index. They investigated EVO oil adulteration with hazelnut, sunflower, and canola oils at 25, 50, and 75% concentration ratios. The ANN model used was the MLP, which proved effective in detecting EVO oil adulteration.

In Kumar (2017), the EVO oil adulteration detection utilizes HPLC as input data, derived from a dataset taken from literature, which consists of 118 samples (71 olive oils, 42 other edible vegetable oils, and 5 mixtures of olive and non-olive oils). The Kohonen map (SOM) was employed as the ANN model for analysis. The results demonstrate that chromatography, combined with this ANN model, achieves an accuracy of about 95% in classifying the oil samples.

In Izquierdo et al. (2020), the study involved 24 samples, including 6 pure EVOOs and 18 mixtures adulterated with different concentrations (2%, 4%, and 8%) of sunflower oil, refined olive oil, and olive pomace oil. The 24 samples were recorded using a thermographic camera during cooling. From the videos, the authors obtained over 20,000 images that served as input for the ANNs. They developed several models based on CNNs, which achieved accuracies ranging from 97% to 100%.

The objective of Bl et al. (2022) is to demonstrate that a voltammetric e-tongue, when integrated with ANN capabilities, can accurately determine the presence of synthetic antioxidants (BHA, BHT, PG) in edible olive oil (EOO). For this purpose, the authors analyzed a commercially available EOOs with added antioxidants, resulting in a total of 52 samples for calibration and 36 for validation. The e-tongue readings were processed using an ANN with three layers. The Relative Error of Prediction Percentage (REP%) for the three substances ranged from 8.3% to 10.3%, favorably comparing with results from previous studies.

In Wu, Du et al. (2024), the authors employ SERS combined with LSTM neural networks to detect and quantify the presence of phthalates in EVOO. For this study, the authors prepared EVOO samples contaminated with six different concentrations of two commonly used phthalates, from which 540 spectra were obtained. Two LSTM models were developed: one for qualitative analysis (classification) and another for quantitative analysis. These models were compared with the performance of RF on the same dataset. Generally, RF outperformed the LSTM models. However, the comparison results were reversed when the models were coupled with a dimension reduction technique, namely

t-Distributed Stochastic Neighbor Embedding (t-SNE). This led to an accuracy of 98% for classification and to $R^2 > 0.97$ for quantification.

The study presented in Wu, Zhang et al. (2024) investigates the capability of three neural networks (CNN-LSTM, AlexNet, ResNet) to differentiate between blended oils composed of olive oil, soybean oil, and sunflower oil in varying proportions. This analysis is based on Raman spectra, with the objective of verifying the accuracy of the blends' labeling. The authors prepared 60 unique blends, from which a total of 2160 spectra were obtained. All the neural network-based models outperformed a traditional chemometric method, namely Partial Least Squares (PLS), with the CNN-LSTM proving to be the most effective, achieving an R^2 prediction value greater than 0.995.

In Zeng et al. (2023), the authors propose to use convolutional neural network based on a new Raman spectra data denoising framework to enhance the prediction accuracy of soybean oil concentration in olive oil.

In Wu et al. (2023), nine types of vegetable oil adulteration were prepared, with ten different adulterant concentrations, ranging from 5% to 50%, in 5% steps, each repeated ten times. Thus, a total of 900 doped samples were collected; additionally, 60 pure oil samples (olive, sesame, soybean, peanut, corn, and palm) were included. The method used for detecting adulteration was 3-D fluorescence spectroscopy, and five machine learning models were tested: k-NN, RF, SVM, PLS, and CNN. At a 5% adulteration concentration, the best-performing model was SVM, with accuracy ranging from 82% to 100% for the different adulteration types; at concentrations of 10%, accuracy consistently reached 100%.

Adulteration has been studied through image analysis in Pradana-Lopez et al. (2022). Five types of Spanish EVO oils were selected, with sunflower and corn oils used as adulterants. More than 302,000 images of oil droplets were used to train and test seven CNNs: one for classifying the EVOOs, one each for detecting and quantifying adulterated samples of the five types of oil, and one capable of performing both of these tasks across all types of oil, effectively dealing with 45 classes. This work is based on the premise that all the different oils exhibit unique fluid-dynamic properties. The authors recorded 30 min videos to capture the evolution and expansion of the droplet; these videos were then divided into frames to form the dataset. Various adulteration amounts were tested, namely 10%, 7.5%, 5%, and 2.5% by weight. The developed CNN models achieved very high performance; in particular, the model capable of both classifying and quantifying adulteration achieved an accuracy of 96.7%.

6.2.2. Classification (CLASS)

In the research papers (Vega-Márquez et al., 2019; Vega-Márquez et al., 2020), the authors conducted an analysis of 701 olive oil samples, comprising 156 LOOs, 355 VOOs, and 190 EVOOs, using GC–IMS. The primary objective was to evaluate the performance of CNN and FNN in classifying the quality of olive oil. Specifically, they focused on three classification categories: EVOO/VOO/LOO, EVOO/non-EVOO, and LOO/non-LOO. In the first study, the CNN model achieved classification accuracies of 83%, 88%, and 85% for these categories, respectively. In the second study, an FNN-based model demonstrated improved accuracies in the latter two classifications (90% and 95%), and was favorably compared with other machine learning techniques such as k-NN and SVM.

The classification in EVOO/VOO/LOO was also the subject in Venturini et al. (2021), where fluorescence spectroscopy was used as data input. 27 olive oil samples were analyzed: 12 EVOOs, 8 VOOs, and 7 LOOs. Eight machine learning algorithms were tested: SVM, Naïve Bayes, multinomial logistic regression, PCA combined with LDA, DT, RF, k-NN, and ANN. The latter, a FNN, had its parameters determined through a grid-search approach. The best-performing configuration had three layers, 32 neurons in each layer, and was trained for 1000 epochs. They showed that DT, RF, k-NN, PCA+LDA (with 30 out of 1024 features), and FNN achieve an accuracy above 99%, and that FNN is the most suitable since it does not require any data pre-processing.

In Chen et al. (2022), the authors employed LIFS to differentiate among twelve oil samples, comprising eleven EVOOs and one Refined Olive Pomace Oil (ROPO), resulting in a total of 72,000 spectra. The study primarily focused on comparing the effectiveness of SVM with that of 1D-CNN. A significant aspect of their work was the development of an optimized model featuring compound convolutional layers, which demonstrated superior performance compared to both the standard 1D-CNN architecture and SVM in discriminating these oil samples.

In Ordukaya and Karlik (2017), 12 different olive oils were collected from various regions in Turkey, and an e-nose was used for classifying them into 12 classes. For each oil, 10 samples were included in the study, totaling 120 samples. Two methods were developed: in the first, all 32 sensor values were used as input, while in the second, the number of features was reduced to 8 through PCA. In both methods, several ML algorithms were applied: Naïve Bayes, k-NN, LDA, DT, SVM, and MLP. The second method outperformed the first with all the ML algorithms, among which Naïve Bayes was the best, although its accuracy is around 70%.

In Rashvand et al. (2023), the authors employ dielectric measurements, specifically phase shift and gain voltages, alongside chemometric quality parameters such as peroxide value, acidity, total phenols, fatty acids, and sterol compounds. This study analyzed 150 oil samples derived from 18 batches of olives, sourced from 3 different cultivars. These olives were collected at three distinct ripening stages and processed using two different mechanical harvesters. The study aimed to assess the effectiveness of various methods in differentiating among the 18 original olive batches. The comparison highlighted MLPs with one and two hidden layers as the most effective methods, surpassing other techniques such as DT, SVM, PLS-DA, and PCA Discriminant Analysis.

In Chien et al. (2019), several vegetable oils were analyzed using a spectrometer for VIS–NIR and SWIR spectroscopies. After preprocessing the spectral data, MLP and SVM were used to determine the type of oil under consideration by classifying it into one of five groups, with virgin olive oils as one group. The study demonstrates that the developed model is effective in accurately predicting the right group of an edible oil.

In Hou et al. (2020), LF-NMR spectroscopies were employed as the data source for the classification of edible oils. Specifically, 52 brands of 11 different edible oils (soybean, rice, olive, walnut, peanut, canola, sunflower, grape seed, linseed, virgin coconut, and corn) were analyzed. For each sample, 9 measures were retrieved (resulting in 468 measures in total). Two models were developed, a 1D-CNN (for signal amplitude) and a 2D-CNN (for images), with their performances compared. The 1D-CNN was found to be the best-performing model.

6.2.3. Cultivar classification (CULT)

In Binetti et al. (2017), cultivar classification is performed by using three datasets comprising four Apulian varieties of olives: one of standard merceological parameters, one obtained from NIR spectroscopy, and one from ^{1}H nuclear magnetic resonance. They analyzed 888 samples of the EVO oils produced in laboratory and obtained in two crop years from 444 plants, whose variety was genetically proved, and 17 oils industrially produced. They used merceological data together with NIR spectroscopy data, and NMR data alone. They developed eight MLP models (four for each of the previous two groups of data), with different initial datasets: the full dataset, a dataset obtained by applying the LASSO algorithm to identify the most significant predictors, a dataset obtained by selecting the variables with standard deviation criterion, and a dataset obtained by selecting the variables with normalized standard deviation criterion. The results showed that the models built on NMR data have higher accuracy (always above 90%) than those built on merceological plus NIR data, thus promoting NMR as a powerful tool for classifying oil cultivars. Moreover, they tested the eight models with the 17 oils industrially produced, and in this case, the accuracies

were lower, although the MLP developed with NMR data after LASSO selection reached an accuracy equal to 88.2%.

In Cervera-Gascó et al. (2023), 124 plots were selected in the region of Castilla-La Mancha (Spain), which has areas assigned to protected designations of origin and quality marks. The predominant varieties in these areas are seven. They developed an MLP model with one hidden layer, named POLIVAR. Then, 22 chemical parameters for each oil were retrieved. To determine whether some parameters were more significant than others, the authors adopted a *brute force* approach by varying the number of parameters injected into the model, starting from 1 parameter. They found that 5 parameters led to accuracy greater than 98%, by checking the accuracies obtained for all the combinations of 5 parameters (26,334 combinations).

Sayago et al. (2019) combined NMR fingerprinting with gas chromatography of the unsaponifiable fraction of the oils to investigate varietal and geographical provenance. The characterization of the unsaponifiable fraction is a traditional method used for traceability and authenticity. The authors selected 49 samples of EVO oils from 17 orchards located in four municipalities of the province of Huelva (Spain) and three cultivars. They applied LDA, SIMCA, PLS-DA, and several machine learning algorithms (RF, SVM, and MLP). They show that the most discriminant variables were tocopherols, squalene, some sterols and aliphatic alcohols, and that NMR resonances related to fatty acid chains displayed significant differences among the study groups. All three developed models showed high accuracy in recognizing the cultivar, and also in classifying the geographical origin, albeit with lower performance.

In Senizza et al. (2023), the authenticity of Taggiasca Ligure EVOO is investigated, with authenticity defined in terms of variety, origin, and blending. The authors selected 408 samples from three consecutive growing seasons and applied UHPLC (Ultra-High-Pressure Liquid Chromatography) coupled with ESI/QTOF-MS (ElectroSpray Quadrupole-Time-Of-Flight hybrid Mass Spectrometry) to profile phenolic compounds and phytosterols. Subsequently, they conducted unsupervised hierarchical analysis to describe metabolomic patterns and analyzed them with orthogonal PLS-DA. They also utilized the variable importance projection (VIP) method to select relevant markers. Furthermore, they developed a FNN with three hidden layers using the abundances of 156 metabolites as inputs. The FNN model reached an accuracy of 98.4%.

6.2.4. Geographical classification (GEO)

The geographical classification of olive oils is critical for product authenticity, including PDO labeling. Given EVO oil's value in the food sector, particularly in Mediterranean countries, it is important to verify proper labeling. In the period covered by this review, 16 papers have been published on this subject.

In Violino et al. (2020), an open-source VIS–NIR spectrophotometer was used. The study involved 67 Italian and 25 foreign EVOOs, with the goal of distinguishing between Italian and foreign oils. MANOVA and ANN models were tested: both models identified the most significant wavelength ranges for origin determination. The ANN, a FNN with one hidden layer, proved to be more accurate than MANOVA, achieving a correct classification percentage of 86.5% on testing samples.

In Wang et al. (2020), five pre-existing datasets (fruits, meats, olive oils, etc.) were utilized to validate a proposed CNN-based method called "ResidualSpectra". The oil dataset consists of 120 FTIR spectra in the range of 799–1897 cm^{-1} with a total of 562 features. The proposed model is a combination of several CNNs connected in cascade. The outcomes of the proposed model were compared with other simpler CNN models and with approaches such as LDA, ANN (unspecified), and SVM. The proposed model outperformed all other compared methods, demonstrating that ResidualSpectra achieves high performance independently of the preprocessing techniques, namely, it can work directly on raw data.

The three works (Lia et al., 2020, 2021, 2022) analyze the same dataset comprised of 30 Maltese and 35 non-Maltese EVOOs samples. In Lia et al. (2020), ^{1}H and ^{13}C NMR fingerprints were used as input data. PCA applied to both ^{1}H and ^{13}C spectra revealed no significant clustering related to geographical origin in the former, while it did in the latter. Subsequently, PLS-DA and FNN were applied comparing several different spectral pretreatments and both models proved effective in classifying the origin of the samples. In Lia et al. (2021), FTIR was used as data input. Several preprocessing techniques were applied and compared to the raw signal, and then flat line regions of the signal were eliminated before statistical analysis. Then, after PCA, which did not reveal any clustering for discriminating Maltese from foreign EVOOs, they applied PLS-DA, Sparse Linear Canonical Discriminant Analysis (SLC-DA), LDA, SVM, and a FFN. They assert that all such methods proved to be highly effective in discriminating Maltese EVOOs. In Lia et al. (2022), fluorescence spectroscopy was used as input data. Several preprocessing techniques and PCA were applied to the raw signal. Then, PLS-DA, ANN, and LDA were applied, and all models were shown to be highly effective in classifying Maltese and non-Maltese EVOO samples. They also showed that spectra obtained at $\delta 10$ nm led to higher model performance.

In Gumus et al. (2018), chromatographic analyses were conducted to classify the origin of different Turkish olive oils. The authors selected 49 olive oils from 6 different areas of Turkey and analyzed sterol, fatty acid methyl esters (FAME), and triacylglycerols (TAG). PCA was first applied to identify the parameters capable of differentiating the locations, and then three monolayer ANNs (one for sterol, one for FAME, and one for TAG) were used to classify olive oils according to their geographical locations. They found that the accuracy values of the ANNs for sterol composition, FAME, and TAG were 85.71%, 83.67%, and 81.63%, respectively. In general, they asserted that sterols and fatty acid contents are effective for clustering olive oils from different regions, whereas TAG does not have significant discriminating power.

A similar study is reported in Gumus et al. (2020), where the same samples were analyzed with a number of chemometric tools, among which MLP. The results showed that the best classification algorithms for geographical discrimination are RF, BayesNet and LogitBoost, while the accuracy of MLP was only 85.7%.

An e-nose was used in Palagano et al. (2020) to extract chromatograms to be analyzed for discriminating between European and extra-European EVOOs and VOOs. The authors selected 210 oil samples from both Europe and extra-EU countries. They defined three categories: "Eu", "extra-EU", and "blends". Data were processed with PLS-DA and MLP, which resulted in better performance in classifying the geographical provenance of the oils.

In Liu et al. (2018), THz spectroscopy was used for its capability to capture information on both intermolecular and low-frequency intramolecular modes of compounds. The authors took 320 samples, 80 for each of the four countries (Australia, Spain, Greece and Italy), and after applying PCA and GA to extract the most significative features in the THz spectra, they evaluated the performance of the following models: LS-SVM, BPNN with one hidden layer, and RF. They found that LS-SVM showed the best prediction accuracy, up to 96.25%, while the BPNN accuracy was 86.25%.

In Harrington (2020), a new classifier is proposed, which is a modification of RBM. To assess the performance of the proposed method, the author selected three existing databases, one of which is related to eight fatty acid concentrations for 572 olive oils from nine regions of Italy. The proposed algorithm outperforms other RBM approaches.

In Torreblanca-Zanca et al. (2019), along with the detection of adulterated samples, the determination of PDO is under investigation. See Section 6.2.1 for further details.

The aim of Yousuff and Babu (2023) is to reduce the dimensionality of spectral data without affecting the performance of chemometric tools. The proposed approach performs the dimensionality reduction in two phases: initially, starting from the raw data, a nearest neighbor

graph is constructed, and the probabilities of the edges are computed; then, a FCNN with four layers is employed to obtain the desired low dimensions. The authors tested the proposed scheme on five datasets, including one related to the EVO oil spectra from Greece, Italy, Portugal, and Spain, and compared the proposed method with four other known dimensionality reduction schemes. Subsequently, they applied RF to the reduced data. The proposed method showed better performance compared to the other dimensionality reduction schemes considered.

In Acquarelli et al. (2017), a new classifier based on CNN is proposed and tested on seven food spectral data datasets, including one related to olive oil. The authors compared the proposed algorithm with PLS-DA, Logistic Regression, and k-NN, and the results indicate that it outperforms them in terms of accuracy.

In Ortenzi et al. (2023), the authors evaluated the performance of two different portable VIS–NIR spectrometers for analyzing 203 EVOO samples. These samples were analyzed using three ANNs for various classifications: EU and Extra-EU, Italian and foreign, and different Italian areas versus foreign (5 classes). All three ANNs featured one hidden layer, with the number of neurons ranging from 10 to 70. The test stage accuracy for the three ANNs was 87.8%, 100%, and 65.9%, respectively.

6.2.5. Attribute prediction (ATTR)

Arnaud et al. (2022), Sperti et al. (2022) and Venturini et al. (2022, 2023) is a series of related works (see also Venturini et al., 2021) that employ fluorescence spectroscopy on the same or related samples (ranging from 18 to 45 different oils) from Spain. Typically, each sample is measured 20 times. The authors use the spectrograms to predict one or more quality indicators of olive oil (acidity, peroxide value, K_{270}, K_{232}, and ethyl esters). In these works, various ANNs (FNN, 1D-CNN, and Conditional CNN) are tested and sometimes compared. In some cases, the accuracy does not exceed 85%, partly due to the small training set.

The objective of Martínez Gila et al. (2021) is to employ an e-nose for predicting the content of fatty acid ethyl esters (FAEE) in olives and their corresponding oils. This parameter is crucial for assessing the quality and determining the commercial classification of virgin olive oils. The study analyzes 84 batches of olives and the respective oils, evaluating the performance of Naïve Bayes, MLP with three layers, and Sequential Minimal Optimization in this context. The aim is to distinguish between samples (both olives and oils) that meet the legal standards for being classified as EVOO based on their FAEE content, and to further categorize the compliant ones into two tiers of excellence. The MLP models achieved classification accuracy ranging from 87.7% to 100%, depending on the specific scenario.

In an earlier related study (Martínez Gila et al., 2020), the authors employed the same e-nose (PEN3) to analyze 82 batches of olives, which were collected at one-day intervals, aiming to predict the classification of the resulting oil as either EVOO or VOO. The study compared three methods: Naïve Bayes, MLP, and PLS-DA. The results indicated that the best classification success rates were achieved by the 3-layer MLP, with 84% accuracy for oil and 90% for olives.

In Gonzalez Viejo and Fuentes (2022) the authors utilize NIR spectroscopy and, separately, an e-nose, to classify EVOO samples spiked with varying levels of rancid oil and to predict the peak area of 16 oil aromas. They used a total of 17 samples, analyzed in triplicate, corresponding to four different EVOO varieties with four levels of added rancid oil (0%, 15%, 25%, 50%) as well as the rancid oil itself. An e-nose, consisting of an array of nine different gas sensors, was also used. The measurements, in triplicate, were then divided into 10 subsections, yielding 510 data points. In the testing stage, the accuracy of the two classifying ANNs was 81.8% and 91.3%, while the two regression ANNs obtained R equal to 0.88 and 0.84. The results obtained with NIR spectroscopy were more accurate than those obtained with the e-nose, which, however, has the advantage of being a less expensive and portable device.

The objective of Violino et al. (2022) was to evaluate the performance of a portable spectroscopic system operating in the VIS–NIR spectrum, coupled with an ANN model, for determining the quality of EVOOs. A total of 203 olive oil samples were assessed using both a VIS–NIR spectrophotometer and standard sensory panel tests. Based on the Sensory Evaluation, the olive oil samples were categorized into two quality classes: “Superior” and “Standard”. The 3-layer FNN model showed a correct classification accuracy of 95% on testing set (99% overall).

The study in Rashvand et al. (2019) investigates the use of the dielectric factor for predicting the acidity and peroxide contents in olive oil, both of which are important indicators of oil quality. The authors conducted their analysis on 30 samples derived from 10 different oils. They compared the predictive performance of Support Vector Regression and MLPs with one or two hidden layers. Their findings indicated that the MLP models achieved slightly better correlation coefficients ($R \geq 0.99$).

The research presented in Sakar et al. (2022) explores the effects of four different olive oil extraction techniques on the oils’ chemical properties, using olives harvested from three distinct locations in Morocco. This investigation employs various multivariate analysis tools. Regarding ANN, the authors mention the use of a 3-layer MLP with ten input nodes to represent fatty acid contents and four output nodes for the extraction methods. However, the paper lacks detailed information such as the accuracy of the MLP model, suggesting it may not have yielded successful results.

The objective of Ghandour et al. (2021) is to automate the classification of olives based on their potential to produce EVOO or regular OO. The authors visited two mills, one specializing in OO and the other in EVOO, and compiled a dataset comprising 1050 images of olive batches. These images were manually segmented to create 11,470 images of individual olives, each labeled according to whether it was used to produce EVOO or OO. The study involved testing various well-known pre-trained CNN architectures employing transfer learning. It was found that ResNet50 outperformed the others, achieving a validation accuracy of 97%.

In Pariente et al. (2018), book images of 190 olives from four varietals, categorized into four quality-based groups (optimal, acceptable, borderline, and unacceptable), were analyzed using PLS-DA and a 3-layer MLP. Three classifications were modeled: (a) optimal vs. others, (b) optimal and acceptable vs. others, and (c) unacceptable vs. others. For classification (a), both PLS-DA and MLP achieved a perfect classification rate. However, for (b) and (c), MLP outperformed PLS-DA achieving higher accuracies of 93% and 90%, respectively. The MLP had 3 input nodes, corresponding to the average values of the RGB color channels. Additionally, the MLP model was applied to classify category (c) using 45 field-taken images, where it correctly identified 85% of the “unacceptable” olives.

Salvucci et al. (2022) proposes a method for assessing olive quality using images captured from a conveyor belt, which are analyzed using CNNs employing the YOLO architecture, designed for real-time object detection. This approach was applied to both table olives and olives used for oil production. For the latter, 1500 RGB images featuring three different olive cultivars were gathered, resulting in a total of 8320 labeled objects. These images were processed to categorize the olives into seven distinct classes, based on color variations and the presence or absence of defects. The researchers tested various YOLO models. Except for the classification of black olives, which was challenging due to the dark background of the conveyor belt, the models achieved an overall accuracy of 98%.

The use of RGB images and CNNs for the rapid classification of olives is also explored in Figorilli et al. (2022). The research involved a dataset of 557 olive images, which were manually classified into five categories. Among these, two categories were designated for defective olives, which need to be removed to produce high-quality oil. The classification accuracy achieved by the model was 90.4% for calibrated

images and 88% for non-calibrated images, demonstrating a relatively high level of precision in olive categorization.

The study presented in Aguilera Puerto et al. (2019) investigates an online system that utilizes real-time imaging on a conveyor belt to classify olives into two categories: those picked up from the ground (low quality) and those harvested directly from the tree (high quality). This classification is conducted by analyzing image histograms from different color spaces and texture values derived from three algorithms. The dataset, initially comprising 3913 features, was reduced to a maximum of 40 using PCA. The authors compared the accuracy of SVM and various configurations of three-layer MLP, both before and after the olives were cleaned. This analysis was based on 6325 images representing 100 batches of olives. The classification success rate achieved was 98.4% using SVM before washing and 98.8% using MLP after cleaning.

The study in Martínez Gila et al. (2022) is centered on predicting olive firmness through multispectral imaging. A set of 58 olives, classified as hard or soft by an expert and measured using a penetrometer, were the focus of the research. The multispectral images of these olives were subjected to analysis using three classification methods: Naïve Bayes, MLP, and a variant of SVM. The most accurate prediction of firmness, achieving a 95% success rate, was obtained by analyzing the images using a 3-layer MLP. In this model, the input layer consists of 25 nodes that represent the 25 infrared channels, the hidden layer comprises 58 nodes, and the output layer has 3 nodes corresponding to the three classes: firm olive, soft olive, and background.

In Mora et al. (2017), the authors employ six MLPs to analyze images of olives and classify them into six categories of ripeness. The goal is to assist agriculturists in determining the optimal time to harvest olives. In this study, 202 images of individual olives were used. Each image was compared to a predetermined color scale, yielding a histogram of 30 values, which were then fed into the MLPs. The percentage of correct classifications across the six stages of ripeness ranged from 90% to 98%.

In a similar study (Alkhasawneh, 2021), the authors use a CFNN to analyze images of olives and classify them into three categories: underripe, ripe, and overripe. For this study, 240 images of individual olives were used. From these images, the RGB components were extracted, and the mean and median of each component, together with their respective SDs, were utilized as inputs for the CFNN. The model developed was able to accurately classify the olives into the three designated categories with an accuracy exceeding 95%.

The aim of Khosravi et al. (2021) is similar to the studies mentioned above. In this study, four ripening stages of two different Iranian olive cultivars were considered. The acquisition of the RGB images was designed to mimic field conditions by capturing photos of olives on the branch, rather than of individual olives. A total of 14,885 sample images were generated by taking random rotations of sub-images from 883 original photos and applying random contrast and brightness adjustments. The images, resized to 256 × 256 pixels, were fed into a CNN that classified the olives into the eight designated classes with an overall accuracy of 91.9%. The results showed improvement when the two cultivars were considered separately.

6.2.6. Oxidative stability (STAB)

The most important parameter to assess the shelf life of the olive oil is the OSI. This parameter is influenced by factors such as oil acidity, packaging, and storage temperature. The works related to this topic making use of ANNs are seven and the inputs used included chemical analyses (3 papers), dielectric (2) and fluorescence (1) spectroscopy, and magnetic resonance (1). The most commonly used model was MPL (4 papers), while the remaining studies used FNN (1), ANFIS (1) and CNN (1).

In Ennouri et al. (2019), to establish the connection between the OSI of EVOOs, the geographical provenance, and the cultivar, fatty acids, phenolic acids, and sterols were analyzed. The EVOOs under investigation come from three harvest years and three cultivars from the center and south of Tunisia. Oxidative stability was estimated with the Rancimat test, used as a reference value. Three MLP networks with one hidden layer, one for each cultivar, were developed, with the output being the OSI. All three MLPs showed good prediction capability ($R^2 > 0.98$). Their performances were also compared with MLR, which was found to be less accurate.

The work in Lastra-Mejias et al. (2019) aims to investigate the effects of storage conditions on the shelf life of three monovarietal EVOOs (Arbequina, Cornicabra, and Picual). Three conditions were considered: exposure to light, the bottle material (plastic or glass), and the clarity (brown or uncolored glass). The paper also explores the adulteration problem by mixing fresh oils with expired ones. The authors prepared a total of 54 samples, 18 of each monovarietal EVOO, and within the 18 samples, 6 were pure fresh oil, whereas the other 12 were mixed with expired EVOO at 1% and 5% concentration. Fluorescence spectroscopy was used as data input, employing six light sources (blue, green, pink, white, UV LEDs, and a UV laser diode), each repeated 10 times for every sample over a 58-day period. Pre-processing was computed to reduce the number of variables by selecting the most relevant signals among the 1912 instances given by spectroscopy. The reduction was performed with the relief-F algorithm, leading to 17 or 18 variables (depending on the EVOO varietal). Then, four MLPs were developed by fine-tuning the parameters, each one having a single hidden layer and a single output neuron. The classifiers obtained accuracies ranging between 91% and 100%.

In Sanaeifar et al. (2018), using dielectric spectroscopy and computer vision, eight values connected with the OSI and the quality in general, are analyzed. The work aims to characterize the olive oil quality indices during storage by developing classifiers and prediction models. A VOO was poured into glass bottles left open in a oven at 60±1 °C. Measurements were taken every 3 days for 24 days, in five copies and averaged. A total of 270 samples were analyzed using dielectric spectroscopy, yielding 384 values related to permittivity. Additionally, from the images, 9 values related to the color features were obtained, resulting in a total of 393 features for each sample. For the storage classifier, the number of features was reduced to 23, comprising 19 dielectric voltages and 4 chromatic components (G, H, L^* and b^*). Three classifiers were developed: MLP, SVM, and Bayesian network, with the latter proving to be the best one. For the quality indexes, only MLP and SVM were developed and compared with MLR, and in this case, SVM proved to be the best one for all indexes.

The same dataset from dielectric spectroscopy was used in Sanaeifar and Jafari (2019) to predict the OSI. The oxidation of the samples was measured 9 times at intervals of 3 days using the Rancimat test. Here, the 393 features for each sample were reduced to 65 (62 dielectric voltages and 3 chromatic components, namely hue, saturation, and b*) using the correlation-based feature selection method, and then utilized as input for three different classifiers: ANN, SVM, and MLR. SVM and ANN were found to be the best predictors of the OSI, with R equal to 0.979 and 0.978, respectively.

The subject of the work in Şahin et al. (2019) is to model the enrichment process of adding olive leaf extract to virgin olive oil for increasing its quality and shelf life. This enrichment has been done by varying the mass of the extract, the time and the amplitude, that are used as input data of the ANN model. The ANN is a MLP with one hidden layer, and one neuron output. A total of 20 experiments were carried out, with 70% of them used for training. A variety of neuron counts in the hidden layer (5, 10, 15, and 20) were tested. Four values related to oxidative stability (total biophenols, oleuropein, total carotenoids, and peroxide value) served as the outputs of the MLPs. The results indicate that the optimal number of neurons in the hidden layer is 10 for all four models tested, leading to an error between the predicted values and the ground truth always less than 3%.

The shelf life of VOOs is studied in Arabameri et al. (2019) as a function of the starting composition of the oil and the storage temperature (25 °C and 37 °C). The authors took 10 VOOs of different

qualities and stored them for 36 months. Analyses were performed at time 0 and at 36 months. The variables checked were fatty acid profile, diacylglycerols, tocopherols, and total phenolic content, measured with gas chromatography, HPLC, and other methods. These variables, measured at time 0, represent the input of the developed model, while the ultraviolet extinction coefficient (K_{268}), directly connected with oxidative stability, is the output. The results show that the proposed model (ANFIS), which is also compared with MLR, predicts the oxidation status with high accuracy.

In Aroca-Santos et al. (2018), the influence of temperature on the properties of EVO oils was studied. Six samples of two Spanish EVO oils (three samples each) were kept at three temperatures (3 °C, 40 °C and room temperature of about 23 °C) and in absence of light. VIS spectroscopy was applied 52 times (over a period of three months) to the four samples maintained at 3 °C and 40 °C. Four absorption peaks (487 nm, 538 nm, 616 nm, and 672 nm) were used as input for the three-layer MLP developed. The model was able to detect the time and the temperature at which the oil had been kept, with a time error between ±6 and ±4 days, while achieving 100% performance in distinguish the storage temperature (3 °C and 40 °C).

6.2.7. Process optimization (OPT)

The study presented in Asgari et al. (2017) utilizes a MLP to model the ultrasound-assisted bleaching of virgin olive oil. Following initial screening tests, the authors conducted a series of experiments to explore various combinations of parameters: ultrasonic power level (0%, 50%, 100%), bleaching clay dosage (0.5%, 1%, 1.5%), process temperature (45, 55, 65 °C), and process time (15, 20, 25, 30 min), resulting in a total of 108 experimental runs. A MLP with two hidden layers was developed to predict changes in Lovibond red (Lr) and peroxide value (PV) during the bleaching process. This model was subsequently employed in a multi-objective optimization procedure using GAs, aiming to identify the parameter combinations that yield optimal Lr and PV values. Compared to conventional industrial practices, the authors reported a 35% reduction in bleaching clay dosage, a 57% reduction in process time, and a 35% reduction in process temperature.

In Funes et al. (2018), the authors utilized a MLP to optimize the clarification process of EVOO, aiming to produce oil with specific desired characteristics. The study involved various assays, which included modifying the oil and water flows and adjusting the temperature during the clarification process through vertical centrifugation. For each of the 35 experiments conducted, NIR spectral samples of oily must and EVOO were collected in triplicate, resulting in a total of 105 oil samples. These samples were analyzed for acidity, peroxide index, UV absorption, free fatty acids, polyphenols, and tocopherols. The authors developed distinct MLP models, each dedicated to predicting one of the oil parameters. These 3-layer MLPs were structured with one output node, from 23 to 32 input nodes, and either 11 or 22 nodes in the hidden layer. The correlation coefficients (R^2) between the estimated values from the models and the actual data were either above or close to 0.9 for most of the parameters, indicating a high level of accuracy in the predictions.

In a previous work (Funes et al., 2017) of similar scope and objectives, the authors utilized 8 FNNs to predict the quality indicators of the oil (including acidity, peroxides, polyphenols, etc.) and the extractability of the process (quantified by pomace fat content and moisture), based on various production parameters and NIR spectrum analysis. In 693 assays, a total of 14 production parameters were modified across different phases, such as olive fruit temperature, sieve diameter, rotational velocity during paste conditioning, water addition, and paste flow in extraction centrifugation, among others. The resulting models achieved $R^2 > 0.9$ in all cases, with the exception of ultraviolet absorption coefficients.

6.3. RQ3 - What are the main types of ANNs that are used during the olive oil processing?

The frequency of use of various ANNs in the primary studies is summarized in Fig. 5 (right part). Some papers compare more than one ANN architecture, while 11 fail to specify the type of ANN used.

Most primary sources utilize ANNs for classification (56), whereas 18 employ them for regression, and 5 papers apply them for both purposes. This follows from the fact that many papers address tasks naturally suited to classification, such as determining the geographical origin of oil, deciding its quality (EVOO vs. VOO), or if it has been adulterated, while the papers using ANNs for regression focus on more quantitative questions, like the oil content of desired or undesired substances.

The primary sources can be categorized into the following four main groups.

Works employing MLP, FNN, or similar ANNs for classification and/or regression. An example of using an MLP for classification is provided in Binetti et al. (2017), where the authors aimed to distinguish between four olive cultivars. They trained and tested various MLPs on a set of 888 samples of the EVOO obtained from 444 genetically verified plants. They used merceological data and the results of NIR and NMR as inputs in various combinations and employed LASSO (Least Absolute Shrinkage and Selection Operator), a regression analysis method, to perform variable selection in order to enhance prediction accuracy. The most effective MLP model was then tested on 17 industrially produced oils, successfully classifying 15 of them.

Conversely, in Sanaeifar and Jafari (2019), MLPs are utilized for regression tasks, particularly for predicting the OSI using dielectric spectra (393 features reduced to 62) and 3 chromatic components. The authors experimented with MLPs having one hidden layer containing varying numbers of neurons (ranging from 1 to 30) and determined that the optimal performance was achieved with a configuration of 25 neurons. The resulting MLP, with a 65-25-1 topology, predicts OSI with an R value of 0.978, whereas the best result found by the authors is $R = 0.979$, obtained with an SVM.

Studies using CNN or 1D-CNN for image or spectral analysis. For example, in Izquierdo et al. (2020), a method for detecting adulteration is presented, which focuses on comparing the thermal behavior of pure and adulterated EVOO. The study involved 24 samples comprising 6 pure EVOOs and 18 mixtures adulterated with different concentrations (2%, 4%, and 8%) of sunflower oil, refined olive oil, and olive pomace oil. The 24 samples were recorded with a thermographic camera while cooling, following heating to 45 °C. From these videos, the authors extracted 22,782 images of 288 × 383 pixels, which were used to train and evaluate a set of CNNs. The final CNN models, each comprising three convolutional layers and two max-pooling layers, achieved accuracies ranging from 97% to 100% in both qualitative assessments (images classified as pure/adulterated EVOO) and quantitative measures (adulterant concentration).

In Venturini et al. (2023), 1D-CNNs are employed to predict five chemical quality indicators of olive oil—acidity, peroxide value, K_{270}, K_{232}, and ethyl esters—using a single fluorescence spectrum obtained with a low-cost portable sensor. This analysis was performed on a dataset of 440 samples collected from 44 different olive oils. The architecture of the 1D-CNN developed in this study includes a convolutional layer, followed by a max-pooling layer, and then a second convolutional layer. This is succeeded by two dense layers and concludes with an output layer featuring a single neuron employing the identity activation function. In general, the results obtained are encouraging. For instance, the authors observe that the 1D-CNN can predict the acidity exceptionally well, except for two LOOs, which have values well above the limit for EVOO and are poorly represented in the samples.

Research involving less common ANN types. For example, in Arabameri et al. (2019), an ANFIS was utilized to predict the oxidative stability of VOO during storage as a function of time, storage temperature,

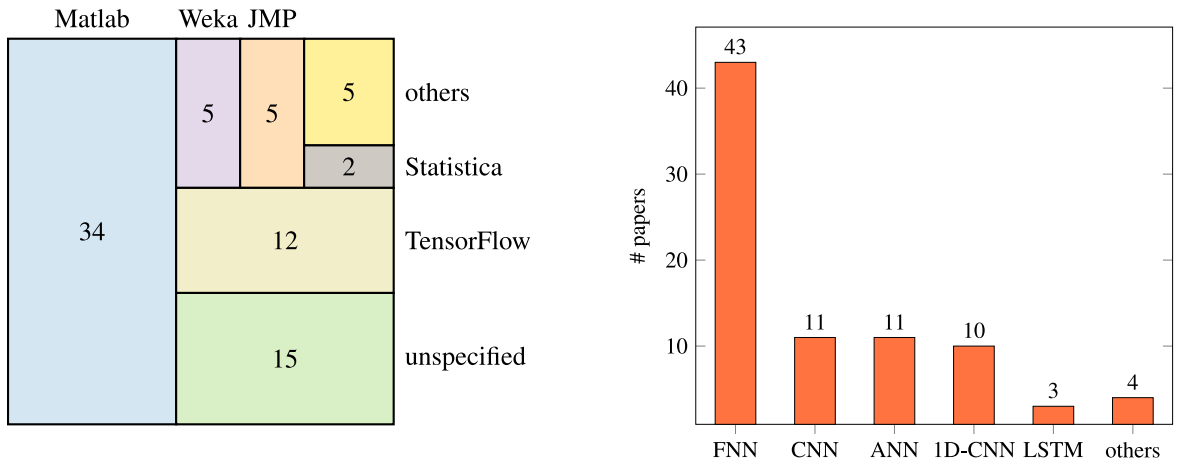


Fig. 5. On the left, the software used in the primary studies for the development and/or application of ANNs. The most commonly used include Matlab (The MathWorks Inc.), TensorFlow (Google), Weka (University of Waikato), JMP (SAS Institute), and Statistica (TIBCO Software Inc.). On the right, the ANN models utilized in the primary studies, where FNN also comprises MLP, BPNN, and FCNN, while the "others" are SOM, ANFIS, RBM, and CFNN.

total polyphenol, α-tocopherol, fatty acid profile, K_{268} coefficient, and diacylglycerol content. In this study, 10 samples of oil were analyzed initially (time 0) and then after 36 months of storage at either 25 °C or 37 °C. According to the authors, ANFIS can predict the quality of oil samples with high accuracy ($R^2 > 0.988$), demonstrating the significant influence of various parameters on oxidative stability.

Papers proposing a variations of a known ANN architecture. In Arnaud et al. (2022), the authors explore the use of a low-cost sensor based on fluorescence spectroscopy for assessing the quality of olive oil (45 oils, each measured 20 times) and distinguishing between EVOO, VOO, and LOO using different ANN models. In particular, the authors propose a variant of the 1D-CNN, termed Conditional CNN, to accommodate the variability in oil characteristics due to different producers or harvest years. They adapt the 1D-CNN architecture by incorporating two input nodes with possible values of (1,0) and (0,1), providing an additional bit of information about the oil. These nodes are not connected to the convolutional layers but directly to the subsequent dense layers. While the achieved accuracy (82%) is not outstanding, it is favorably compared with those from standard 1D-CNN and FNN models on the same dataset.

As depicted in Fig. 5 (on the left), most papers utilized Matlab (usually with its Deep Learning Toolbox or Neural Networks Toolbox), while several that focused on convolutional neural networks did so with the aid of TensorFlow, often in conjunction with the Keras library. Similar to the situation with neural network architectures, many papers did not explicitly mention which software, library, or programming language was employed.

Hyperparameters are crucial elements in the design and training of neural networks that are set before the learning process begins and are not learned from the data. They control various aspects of the training process, impacting the performance and efficiency of the neural network. Beside the network architecture itself (number of layers, number of neurons per layer) the most commonly reported hyperparameters in the 79 primary studies are:

Optimizer: The algorithm used to adjust the model's weights, explicitly mentioned by 43 papers. The most cited optimizers were Adam and its variants (13 times), Levenberg–Marquardt algorithm (13 times), and SGD and its variants (11 times).

Activation functions: Functions applied to each neuron's output to introduce non-linearity, reported by 42 studies. The most common were ReLU (17 times), sigmoid (13 times), and tansig (10 times).

Number of epochs: The number of times the entire training dataset passes through the network, and *number of iterations*, the number of updates made to the model's parameters during training, reported by 26 and 14 primary studies, respectively. Note that some authors might use "number of iterations" to refer to the "number of epochs" or vice versa.

Mini-batch or batch size: the number of training samples used to compute the gradient in each iteration, cited by 11 papers.

Among the 79 primary studies, 18 do not provide any information about the hyperparameters, apart from the size of the neural network itself.

6.4. RQ4: What metrics are adopted to evaluate the ANN model performance?

The effective performance assessment of the ANN model is a key aspect for the selection of the best ANN. Fig. 6 shows the metrics used across the primary studies. We refer to Section 2.3 for the description of these metrics. Many primary studies use more than one metric. The most commonly used metric to evaluate performance of an ANN is classification accuracy (35 times explicitly mentioned + 22 times inferred by a confusion matrix), representing how well a model performs in correctly labeling a data instance. Its large adoption in the olive oil sector is due to its simplicity and ease of interpretation. It has been largely used for predicting adulteration (Torreblanca-Zanca et al., 2019), for classifying the quality of olive oil according to the three classification categories: EVOO/VOO/LOO, EVOO/non-EVOO, and LOO/non-LOO (Vega-Márquez et al., 2020), for geographical discrimination (Violino et al., 2020), for cultivar classification (Binetti et al., 2017). Another less commonly adopted metric in the olive oil sector is R or R^2 (16 times), a general statistical measure able to assess the goodness-of-fit of a regression ANN model. In many primary studies (Ennouri et al., 2019), the exhibited R^2 value is close to 1, suggesting that the model accurately captures the underlying relationship in the dataset, with predictions that closely align with the actual values. Other evaluation metrics such as MAE (8 times) (Torreblanca-Zanca et al., 2019), RMSE/RMSEC (8 times) (Sanaeifar & Jafari, 2019), MSE (3 times) (Asgari et al., 2017), are also used alongside or instead of R^2 when evaluating ANN models. Finally, the Kappa value, another metric that is very rarely used (Martínez Gila et al., 2021), represents the level of agreement between the classes predicted by the ANN model and the true classes in the dataset.

7. Key findings, challenges and future research directions

The results of our review demonstrate ANNs can outperform standard chemometric methods in analyzing complex data, such as those encountered in the study of olive oil, due to several inherent advantages. For instance, ANNs are adept at modeling non-linear relationships without the need for pre-specifying the form of the relationship,

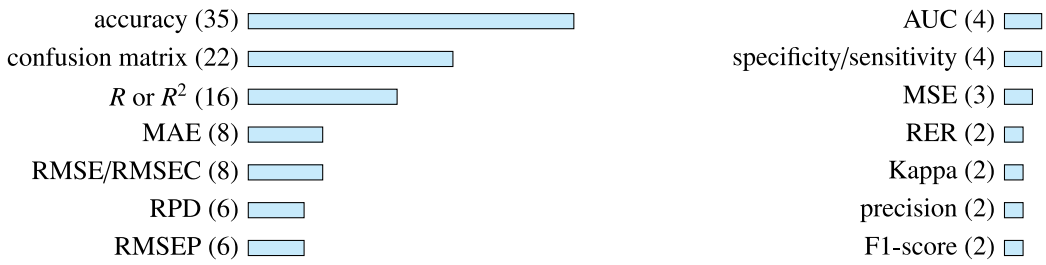


Fig. 6. Metrics employed multiple times across primary studies. For studies that present a confusion matrix for classification results, accuracy, precision, specificity and sensitivity are also inferred, as a confusion matrix implicitly provides these metrics.

which is a common requirement in many chemometric techniques; ANNs can easily integrate diverse types of data (spectral, chemical, sensory, etc.) without needing extensive preprocessing or transformation; ANNs have shown resilience in dealing with noisy or incomplete datasets, which are common in real-world applications.

On the other hand, ANNs are often considered "black boxes" due to their complex structure, making it difficult to interpret how they make predictions and to understand the factors influencing their output. In response to this challenge, Explainable Artificial Intelligence (XAI) (Ali et al., 2023) has emerged as a crucial research direction aimed at mitigating this problem. XAI strives to develop methods and techniques that can provide clear explanations and insights into how specific inputs lead to particular outputs.

Emerging techniques like transformer-based architectures and Graph Neural Networks (GNNs) have gained popularity in various fields due to their advanced capabilities in handling sequential data and complex graph structures. However, finding applications for them in the field of olive oil characterization and production presents a challenge, since the field specific requirements do not align well with the primary strengths of these advanced neural network models. In particular, transformers require large amounts of data to train effectively and are primarily designed for sequential data, such as text or time series. Based on the primary studies we have analyzed, it appears that the available datasets may not be large enough to fully leverage the capabilities of transformer architectures. As for GNNs, they are designed to work with data that can be represented as graphs, where relationships between entities are crucial. In olive oil characterization and production, the data is typically tabular and do not naturally form a graph structure.

Another key challenge in applying ANNs to olive oil processing lies in obtaining large and diverse datasets. This difficulty arises from factors such as seasonality, regional variations, and the human and financial costs associated with sample collection and analysis.

Regarding possible bias in the input data, we can distinguish between two typical scenarios. In most papers dealing with oil adulteration, authors generate the samples to be analyzed by mixing pure oils and adulterants in known proportions, resulting in an inherently balanced input dataset across various classes. On the other hand, this may not be feasible in other cases where oils or olive samples are collected from multiple sources based on availability and opportunity. For example, in Sayago et al. (2019), the 49 olive samples belong to three different varieties, with counts of 21, 20, and 8. In some cases, such as Martínez Gila et al. (2021), where the authors perform a binary classification, the class distribution is determined only after a set of analyses is conducted on the collected samples.

As showed in Table 8, approximately 86% of the primary studies use a dataset with fewer than 1000 samples, which may not be enough to represent the variability in the data and to train the model effectively, leading to typical issues such as overfitting, model instability, high bias or high variance in the model's predictions, as well as a lack of statistical significance of the results. To mitigate these issues, strategies such as data augmentation, transfer learning from related domains, and careful cross-validation can be employed. Additionally, collecting more data or combining datasets from different sources can alleviate the effects of small sample sizes. Furthermore, integrating the results from ANNs with existing knowledge in food science and olive oil production needs interdisciplinary expertise and/or collaboration. The findings of our study suggest that bridging the gap between computational insights and practical applications in olive oil research remains challenging. Avoiding potential issues, such as overfitting, demands a comprehensive understanding of the capabilities and limitations of ANNs.

As reported in Table 8, less than one-third of the primary studies explicitly address or cite the issue of overfitting during model training. Among those that do, the main methods used to avoid or mitigate the problem are L2 regularization (Hou et al., 2020; Wu, Zhang et al., 2024), early stopping (Arnaud et al., 2022; Binetti et al., 2017), and cross-validation techniques such as k-fold, hold back, and excluded rows (Lia et al., 2020, 2021, 2022).

There is not a one-to-one mapping of ANN types to different objectives, which means different types of ANNs are used for the same goal. However, the specific characteristics of FNNs and CNNs suggest that certain ANNs are more frequently used with specific sources of features. For example, FNNs are often adopted for data series or other one-dimensional datasets (Gharehchopogh, Khaze, & Maleki, 2015), while CNNs are more suitable for RGB images (Alsajri & Hacimahmud, 2023; Nafea, Alameri, Majeed, Khalaf, & AL-Ani, 2024; Özbay, Özbay, & Gharehchopogh, 2024), though they also find applications in other domains (Asgharzadeh, Ghaffari, Masdari, & Soleimanian Gharehchopogh, 2023). This inherently influences the choice of ANN architecture in relation to the goal, whether it be adulteration detection or assessment of olive fruits. For instance, Tables 4 and 5 indicates that MLPs are the most commonly used ANNs for establishing the oxidative stability of olive oil, whereas both MLPs and CNNs (including 1D-CNNs) are utilized for attribute prediction, i.e., for predicting characteristics of the oil such as the amounts of polyphenols or peroxides.

We observe that spectroscopy represents a valuable data source, partly because it can be tailored to specific analytical goals by modifying the frequency at which it operates. Spectroscopic techniques, such as NIR or FTIR spectroscopy, can provide rapid and non-destructive measurements that, in conjunction with ANNs, allow for the prediction of various chemical and physical properties of olive oil. Fluorescence spectroscopy, in particular, is popular since it is cost-effective, rapid, and still accurate, especially when employed with portable devices. Additionally, it often does not require pre-processing when its results are input into ANNs. As shown in Table 6, spectroscopy is the preferred technique for various goals, and especially for detecting the presence of adulterants.

Olive oil is a complex mixture with numerous compounds contributing to its characteristics. The diversity of olive oil samples and production methods can make it difficult for a single model to perform consistently well across different types, qualities, or origins of olive oil. This aspect particularly complicates the evaluation of how extensively the results obtained in one study can be generalized to slightly different scenarios and datasets. In general, we observed variation in the metrics used to assess the performance of ANNs in olive oil processing, as

Table 8
In the first column, we report the largest number of data instances (training + validation + test) used in each primary study. The symbol '?' denotes studies for which, to the best of our understanding, we were unable to ascertain the actual number of data instances used to train and test the ANN(s). A check mark in the last column identifies the primary studies that explicitly mention overfitting.

#data	Study	ov.	#data	Study	ov.	#data	Study	ov.	#data	Study	ov.	#data	Study	ov.
302 387	Pradana-Lopez et al. (2022)	✓	693	Funes et al. (2017)		360	Wu, Du et al. (2024)		168	Zarezadeh, Aboonajmi, Ghasemi Varnamkhasti (2021)		82	Martínez Gila et al. (2020)	
72 000	Chen et al. (2022)	✓	600	Liu et al. (2021)		320	Liu et al. (2018)		160	Xu et al. (2018)	✓	79	Chien et al. (2019)	
22 782	Izquierdo et al. (2020)		572[b]	Harrington (2020)	✓	294	Zarezadeh et al. (2023)		150	Rashvand et al. (2023)		65	Lia et al. (2022)	✓
11 470	Ghandour et al. (2021)		557	Figorilli et al. (2022)		294	Zarezadeh, Aboonajmi, Varnamkhasti et al. (2021)		141	Aroca-Santos et al. (2019)		65	Lia et al. (2021)	✓
8 352	Martínez Gila et al. (2022)		540	Lastra-Mejias et al. (2019)		270	Sanaeifar and Jafari (2019)		124	Cervera-Gascó et al. (2023)	✓	65	Lia et al. (2020)	✓
6 325	Aguilera Puerto et al. (2019)		540	Venturini et al. (2021)		270	Sanaeifar et al. (2018)		120[b]	Wang et al. (2020)	✓	60[b]	Yousuff and Babu (2023)	
4 415	Khosravi et al. (2021)		528	Omwange et al. (2022)		254	Torreblanca-Zanca et al. (2019)		120[b]	Acquarelli et al. (2017)		49	Gumus et al. (2020)	
2 160	Wu, Zhang et al. (2024)	✓	510	Gonzalez Viejo and Fuentes (2022)	✓	252	Hou et al. (2021)	✓	120	Ordukaya and Karlik (2017)		49	Sayago et al. (2019)	
1 777	Surya and Senthilselvi (2022)	✓	468	Hou et al. (2020)	✓	240	Alkhasawneh (2021)		118[b]	Kumar (2017)		49	Gumus et al. (2018)	
1 661 ?	Lia et al. (2018)		440	Venturini et al. (2023)	✓	235[a]	Pariente et al. (2018)		108	Asgari et al. (2017)		30	Arabameri et al. (2019)	
1 500	Salvucci et al. (2022)		440	Venturini et al. (2022)	✓	210	Palagano et al. (2020)		105	Funes et al. (2018)		30	Rashvand et al. (2019)	
960	Wu et al. (2023)		440	Sperti et al. (2022)	✓	208	Aroca-Santos et al. (2018)	✓	96	Violino et al. (2021)		25	Chuma and Rasmussen (2022)	✓
900	Arnaud et al. (2022)	✓	420	Yakar and Karadağ (2022)		203	Violino et al. (2022)		92	Violino et al. (2020)		20	Şahin et al. (2019)	
888	Binetti et al. (2017)	✓	420	Zeng et al. (2023)		203	Ortenzi et al. (2023)		90	Capar et al. (2019)		?	Ennouri et al. (2019)	
701	Vega-Márquez et al. (2020)		408	Senizza et al. (2023)		202	Mora et al. (2017)		88	Bl et al. (2022)		?	Sakar et al. (2022)	
701	Vega-Márquez et al. (2019)		375	Soltani et al. (2019)		180	Meng et al. (2023)		84	Martínez Gila et al. (2021)				

[a] Denote studies that use data from other sources: in part.
[b] Denote studies that use data from other sources: in total.

depicted in Fig. 6. This variation naturally stems from the dual nature (classification and regression) of ANNs. However, it prevents a fair comparison among the different types of ANNs and their evaluations. A potential future research direction to address this problem is the development of an assessment framework for benchmarking the performance of different types of ANNs. The goal of such a framework should be also the fair evaluation of the results achieved with the different types of ANNs taking into account the bias that could affect datasets or models. Similar studies have been conducted in other domains, such as medicine (Meng, Trinh, Xu, Enouen, & Liu, 2022). In this context, developing a tool that facilitates the sharing of datasets and trained models could be useful, allowing for easier and fairer comparison of different ANN architectures.

8. Conclusions and future work

ANNs have emerged as exceptionally powerful tools in various fields, including agronomy, where they offer innovative approaches to analyze complex datasets.

In this paper, we proposed an up-to-date systematic review of the literature on the use of ANNs in olive oil production and characterization. Our systematic search on 6 different digital libraries, complemented by snowballing on Google Scholar, led us to select 79 primary studies that have been analyzed to identify objectives, types of ANN, their inputs, and the adopted performance metrics.

Despite the structured and rigorous nature of our review, certain limitations must be acknowledged. The presented classification might not include all the dimensions that could be relevant in the field of ANN applied to olive oil sector; we tried to address this issue using the data of the primary studies for defining the main dimensions of our classification. The choice of the search query as well as of the digital libraries for performing the automated search may have influenced the selection and classification of the primary studies. However, this risk has been mitigated by performing backward and forward snowballing over Google Scholar to find other primary studies.

One of the key results of our review is the identification of major challenges in integrating ANNs into the olive oil sector. While the advantages of these models are evident, several obstacles hinder their widespread adoption. A primary issue is the limited availability of large, high-quality datasets, which restricts the training and generalization capabilities of ANNs. Moreover, the variability in performance metrics across studies complicates direct comparisons and benchmarking, making it difficult to establish standardized evaluation frameworks.

The results of our review show that integrating ANNs into scientific research of olive oil production is undeniably positive, reflecting the growing recognition of their potential to unlock new insights and improve predictive accuracy beyond what traditional statistical methods can achieve. Unlike well-established techniques such as PCA or

MLR, which have become standard tools for experimental scientists, ANNs are not universally acknowledged as ready-made solutions. Their effective application depends on careful design, training, and validation to avoid common pitfalls such as overfitting or underfitting. Missteps in model selection, feature engineering, or hyperparameter optimization can lead to misleading conclusions, reducing the reliability of ANN-driven insights.

To overcome these challenges, it is crucial for researchers to build a solid foundation in machine learning principles and best practices. Addressing data scarcity through collaborative dataset sharing, standardizing performance assessment methods, and promoting interdisciplinary cooperation between agronomists and AI specialists will be key steps toward unlocking the full potential of ANNs in olive oil research. The ultimate goal is to transition ANNs from being perceived as specialized, complex tools to becoming indispensable assets in the scientific study of olive oil production and quality assessment.

This systematic review focused on olive oil production and post-production analyses. For future work, we plan to review and analyze the literature on olive tree cultivation, where integrating ANNs with IoT real-time monitoring could help optimize yield and quality. Further interesting aspects could also include sustainability assessments, climate change adaptation strategies, and the economic impact of emerging technologies in olive farming. Additionally, we aim to explore gray literature sources to capture insights from industry reports, technical documents, and non-peer-reviewed studies.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to check grammar and improve fluency. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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